PCCP

PERSPECTIVE

View Article Online
View Journal | View Issue

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Cite this: *Phys. Chem. Chem. Phys.*, 2017, **19**, 2636

Ionic liquids and deep eutectic solvents for lignocellulosic biomass fractionation

Dannie J. G. P. van Osch,*[a] Laura J. B. M. Kollau,[a] Adriaan van den Bruinhorst,[ab] Sari Asikainen,[c] Marisa A. A. Rocha†[b] and Maaike C. Kroon*[bd]

Received 2nd November 2016,
Accepted 2nd December 2016

DOI: 10.1039/c6cp07499e

www.rsc.org/pccp

Lignocellulosic biomass has gained extensive research interest due to its potential as a renewable resource, which has the ability to overtake oil-based resources. However, this is only possible if the fractionation of lignocellulosic biomass into its constituents, cellulose, lignin and hemicellulose, can be conducted more efficiently than is possible with the current processes. This article summarizes the currently most commonly used processes and reviews the fractionation with innovative solvents, such as ionic liquids and deep eutectic solvents. In addition, future challenges for the use of these innovative solvents will be addressed.

[a] *Laboratory of Physical Chemistry, Department of Chemical Engineering and Chemistry and Institute for Complex Molecular Systems, Eindhoven University of Technology, P.O. Box 513, 5600 MB Eindhoven, The Netherlands. E-mail: d.v.osch@tue.nl*
[b] *Separation Technology Group, Department of Chemical Engineering and Chemistry, Eindhoven University of Technology, P.O. Box 513, 5600 MB, Eindhoven, The Netherlands. E-mail: m.c.kroon@tue.nl*
[c] *VTT Technical Research Centre of Finland Ltd, P.O. Box 1000, FI-02044 VTT, Finland*
[d] *The Petroleum Institute, Dept. of Chemical Engineering, P.O. Box 2533, Abu Dhabi, United Arab Emirates*
† Current address: NALCO Water, An Ecolab Company, Ir.G.Tjalmaweg 1, 2342 BV Oegstgeest, The Netherlands.

1. Introduction

Lignocellulosic biomass can be described as the primary building block of plant cell walls consisting of three main components, *i.e.* cellulose, hemicellulose and lignin. Next to these three main constituents lignocellulosic biomass is composed of small amounts of pectin, proteins, extractives and ash.[1] Per year approximately 2×10^{11} tonnes of lignocellulosic biomass is produced.[2] Plant biomass is produced *via* a photosynthesis reaction between carbon dioxide (CO_2), water (H_2O) and light, in which solar energy is mainly



Dannie J. G. P. van Osch

Dannie van Osch received his MSc in chemical engineering, specialized in process engineering, from Eindhoven University of Technology with honors in 2014. In 2013 he was a finalist in the VSNU Student Research Conference and he obtained a 4th place at the annual Huntsman Design Contest (ECCE9-ECAB2). In 2014 he started his PhD, in which he and his colleagues were the first to publish hydrophobic deep eutectic solvents (DESs) and their use in removal of metal ions from non-buffered water. His research interests are in wood valorization with innovative solvents and the removal of contaminants from water with hydrophobic DESs.



Laura J. B. M. Kollau

Laura Kollau is a PhD student. She holds a Bachelor's degree in Chemistry from Zuyd University and a Master's degree in Chemical Engineering from Eindhoven University of Technology. For her master thesis she investigated transition-metal catalysed reactions in microflow using highly reactive diazonium salts as intermediates. After completing her master's degree she started her doctoral work in Eindhoven in the Physical Chemistry research group of Prof. Remco Tuinier. Her research focuses on the delignification of woody biomass and pulp with deep eutectic solvents and using these innovative solvents to design a new pulping process.

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

converted into chemical bonds and to a lesser extent into chemical energy (only 1%).[3] Lignocellulosic biomass is stated as the Earth's biggest natural resource available.[4] This makes it interesting as a renewable resource, decreasing the dependency of humanity on oil-based resources. Especially the fractionation of lignocellulosic biomass into its constituents – cellulose (40–80%), hemicellulose (15–30%) and lignin (10–25%) – is of major importance for the future use of lignocellulosic biomass as a renewable resource.[5] Currently, the production of paper from wood is the most optimal operational commercial fractionation process, but it uses only 35% of its total biomass feed.[6] The other 65% is considered as waste and in most cases incinerated. Thus, the fractionation should be conducted more efficiently to compete with our oil-based resources. In this review, current fractionation processes are compared to treatments with novel solvent systems such as ionic liquids (ILs) and deep eutectic solvents (DESs). Furthermore, the solubility of lignocellulosic biomass and its constituents is evaluated in these novel solvent systems.



Adriaan van den. Bruinhorst

Adriaan van den Bruinhorst received his MSc in chemical engineering with honours in 2012 at TU/e. His master graduation project titled "Processing Lignocellulosic Biomass with Natural Low Transition Temperature Mixtures (LTTMs)" was awarded the Taminco Green Footsteps Award 2012 in the field of green solvents. After his master's degree, he continued his work at a PhD position at the TU/e. He studied several extraction processes, including metal and volatile fatty acid extraction. Besides these applications, he focused on the fundamentals of DESs formation and elucidated their often complicated phase behaviour.



Sari Asikainen

Sari Asikainen received her DSc (Tech.) in the field of bioproduct technology from Aalto University, School of Chemical Technology. She has over 20 years experience in chemical pulping and bleaching, fibre fractionation and refining.



Marisa A. A. Rocha

Marisa A. A. Rocha was born in 1983 in Gondomar, Portugal. She obtained her PhD in Chemistry from the University of Porto in 2013. She then moved to Eindhoven University of Technology, as a Postdoctoral researcher at Separation Technology Group, where she worked on the design of new solvents (ionic liquids and deep eutectic solvents) to be applied in separation processes. In 2015, she started a Postdoctoral position at Engineering Thermodynamics Group in the University of Bremen, where she continued developing research in the same field. Currently, she is Principal Chemist at Paper Research division at Nalco Water (an Ecolab company), Oegstgeest, The Netherlands. Her research interests are focused on the molecular understanding of the behavior of pure liquids and mixtures based on their physical chemical properties.



Maaike C. Kroon

Maaike Kroon received her MSc (2004) and PhD (2006) in chemical engineering from TU Delft, both with honors. She was visiting assistant professor at MATGAS in Barcelona (2007) and at Stanford University in Palo Alto (2008–2009). At the age of 29, she was appointed full professor at TU/e (2010). Since 2016, she has been working as full professor at the Petroleum Institute in Abu Dhabi. Her research in the field of highly efficient and selective affinity separation processes has been published in over 100 articles and awarded various prizes, including the New Scientist Research Talent award 2015.

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

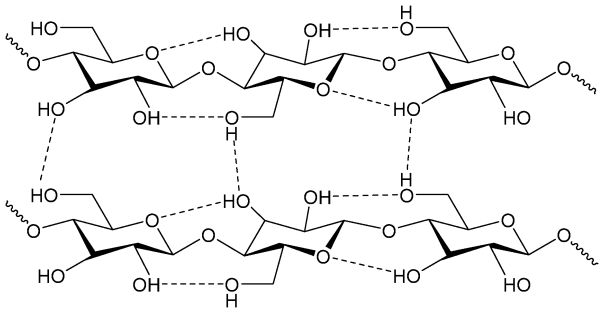


Fig. 1 General form of the biopolymer cellulose with its intramolecular and intermolecular hydrogen bonds between the glucose monomers.

2. Lignocellulosic biomass constituents

2.1. Cellulose

Cellulose is the most abundant biopolymer on Earth.[7] It is a polysaccharide consisting of linear chains of 10 000 or more D-glucose monomers linked together *via* 1–4-β glycosidic bonds.[8] Between these linear chains there are many parallel and anti-parallel hydrogen bonding interactions, which are graphically depicted in Fig. 1.[7] These bound linear chains are packed into microfibrils.[9] The exact nature of microfibrils varies between the different types of lignocellulosic biomass.[10] It can be concluded that the combination of the hydrogen bonding network and the microfibrils gives cellulose its rigid form and recalcitrance to dissolution. The crystalline structure of cellulose can be divided into seven different types, *e.g.* cellulose I (I_α and I_β), cellulose II, cellulose III (III_1 and III_{11}) and cellulose IV (IV_1 and IV_{11}).[10] The main differences between these distinct forms are their variations in the unit cell.

Cellulose has different applications. Probably the oldest and most common application is its use in the paper and pulp industry. Furthermore, it is often used as dissolving cellulose. Next to the use of cellulose itself, its hydrolysis is also of great interest. Multiple publications show the hydrolysis of cellulose into glucose,[11,12] and further decomposition or conversion into ethanol,[13,14] 5-hydroxymethylfurfural[15,16] and other constituents. It is generally speculated that a nearly limitless variety of chemicals can be produced *via* hydrolysis and subsequent conversion of cellulose.[9]

2.2. Hemicellulose

Hemicellulose, like cellulose, consists of groups of polysaccharides, but has a much lower degree of polymerization (100–200). Moreover, hemicellulose does not merely consist of glucose monomers, but it is a branched heteropolymer composed of three different monomer classes, *i.e.* pentoses, hexoses and sugar acids (Fig. 2). The most common pentoses in hemicellulose are D-xylose and D-arabinose. Frequently occurring hexoses are D-mannose, D-glucose and D-galactose. Most known sugar acids are 4-*O*-methyl-D-glucuronic acid, D-galacturonic acid and D-glucuronic acid. The more rare sugars are L-rhamnose, L-fucose, and various methylated neutral sugars.[17,18]

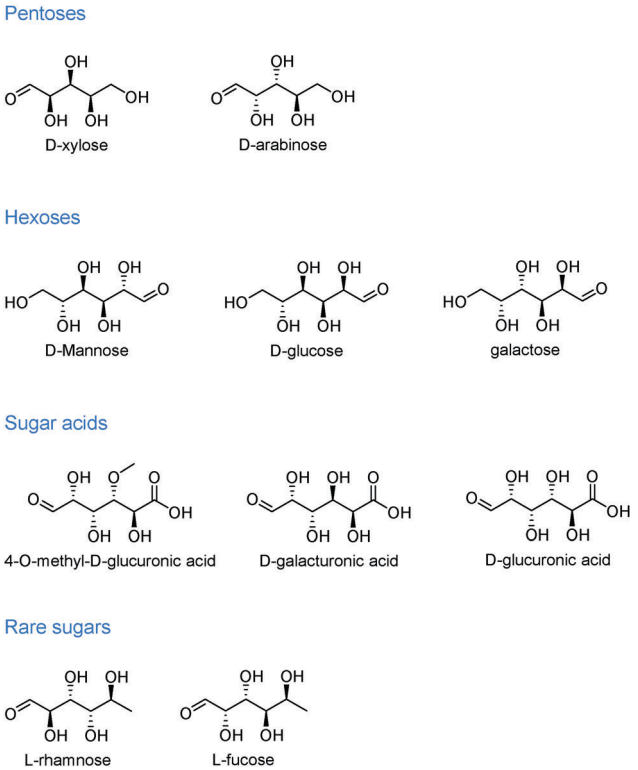


Fig. 2 Monomers of hemicellulose.

It should also be noted that the composition of hemicellulose differs per type of lignocellulosic biomass.[8] Similarly to cellulose, hemicellulose can be hydrolysed and further converted into different products, such as ethanol, xylitol, 1,2-butanediol and lactic acid, which makes further research into the applications of hemicellulose interesting.[18]

2.3. Lignin

Lignin is a three-dimensional amorphous biopolymer insoluble in water in its native form.[19] It acts as a resin between the carbohydrate polymers cellulose and hemicellulose holding the lignocellulosic matrix together.[19,39] Crosslinks between lignin and the carbohydrate polymers increase the mechanical strength and rigidity conferred on woody tissues, which allows trees to grow up to more than 100 meters in height. Moreover, lignin plays a part in the defence mechanism of lignocellulosic biomass. Due to the insolubility and complexity of the lignin polymer, it is resistant to degradation by most microorganisms.[8,20]

Despite investigations on the chemical structure of lignin for more than a century, the exact structure is still under debate.[21] This can be explained by the lack of precise qualitative analytical methods. Especially when the analytical method requires a soluble lignin derived preparation, serious problems occur due to the lack of a method for the isolation of lignin without changing its functional groups.[22] Recently, new methods were introduced showing great promise for the detection of lignin in plant and wood species. These methods are based on nuclear magnetic resonance (NMR) and Fourier transform infrared (FTIR) spectroscopy.[23–25]

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

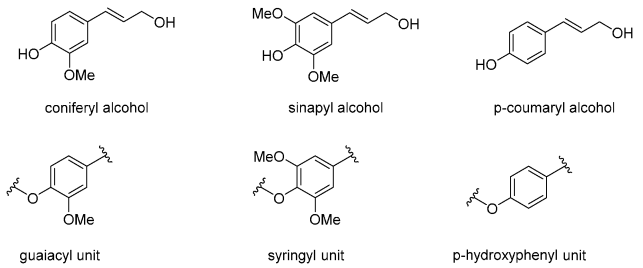


Fig. 3 Monolignols present in lignin and their respective units within the lignin polymer.

Although a suitable analysis method for the characterization of chemical structure of lignin is not available yet, it is widely accepted that lignin consists of the monolignols coniferyl alcohol, sinapyl alcohol, and *p*-coumaryl alcohol, also known as 4-hydroxycinnamyl alcohol, which subsequently after incorporation into the lignin polymer are renamed as guaiacyl, syringyl and *p*-hydroxyphenyl units (Fig. 3).[8,26]

Multiple valuable products can be obtained from lignin.[19] Currently, the most successful commercial valorisation of a lignin derivative is the production of vanillin from the lignosulfonates produced with the sulphite pulping process.[2] Lignosulfonates have also been commercially applied as dispersants (*e.g.* in the construction industry), emulsifiers (*e.g.* in asphalt), and in resins on a smaller scale. Probably, numerous new or easier produced derivatives can be obtained after lignin is successfully removed from the supramolecular structure with cellulose and hemicellulose in lignocellulosic biomass. For example, aromatic compounds such as benzene, xylene, toluene and phenols could be potentially produced.

2.4. Other constituents

Next to the three main constituents, lignocellulosic biomass also contains other components of which the most prominent is water. Since wood and most other lignocellulosic biomasses are hygroscopic, the water content varies with relative humidity of its surroundings. For instance, for raw loblolly pine the equilibrium moisture content typically varies between 10 and 25 wt% at common relative humidity and ambient temperature.[27] The dry material also contains monomeric sugars, organic acids, salts, amino acids, proteins, oils, resins and other hydrocarbons, including flavonoids, colorants, anti-oxidants and fragrances.

3. Conventional processes used for chemical pulping and cellulose dissolution

This section focuses on the most often applied and/or investigated processes for the delignification of lignocellulosic biomass (Kraft, sulphite and organosolv pulping) and production of man-made cellulosic fibres (viscose and Lyocell process).

3.1. Kraft pulping

The origin of kraft pulping, also known as sulphate pulping, dates back to 1870 when a patent was filed for the delignification of lignocellulosic biomass with sodium hydroxide (NaOH) and sodium sulphide (Na_2S). In 1879 it was discovered that the addition of sodium sulphate (Na_2SO_4) during the recovery resulted in lower sulphide losses.[28] This discovery was the basis for the first industrial Kraft process for the removal of lignin and hemicellulose from lignocellulosic biomass, leaving behind cellulose fibres.[29,30]

Nowadays, the kraft pulping process is used in 80% of the world production of pulp. The wood chips are cooked using a mixture of white liquor and spent black liquor at a temperature of 150–180 °C. At this temperature the main part of the lignin is dissolved and also some of the hemicelluloses are removed. The residual lignin, although its content after the cooking is only 2–5% of the dry pulp weight, is intensely coloured. The residual lignin is difficult to remove in the Kraft cook without severe carbohydrate degradation and it is therefore modified or removed in several bleaching stages.

The spent liquor from the cooking is concentrated and incinerated in a reductive recovery boiler, which has two main functions. One is to burn the dissolved organic material and produce a molten salt comprising sodium carbonate (Na_2CO_3) and Na_2S. The other purpose is to recover the heat from the hot flue gases as high-pressure steam for power generation in a turbogenerator system and to meet the steam needs of the mill.[31]

Disadvantages of the kraft pulping process include the need for alkaline chemicals resulting in metal corrosion problems and malodorous smells, and the fact that the obtained lignin contains sulphur, which increases the difficulty for valorisation and requires a higher need for bleaching.[31]

3.2. Sulphite pulping

The sulphite process comprises four different methods for the removal of lignin and part of the hemicelluloses: the acid bisulphite (pH 1–2), bisulfite, *i.e.* magnefite (pH 3–5), neutral (pH 6–9) and alkaline (pH 10–13.5) sulphite process. Cooking temperatures vary from 120 °C to 180 °C, depending on the sulphite process. In comparison to the Kraft process, there are two different methods for the recovery of the chemicals. One method increases the concentration of the weak black liquor *via* evaporation. This concentrated liquid is incinerated in a recovery boiler to produce energy for steam and power generation where magnesium oxide (MgO) and sulphur dioxide (SO_2) are produced. MgO is separated from the flue gases *via* electrostatic precipitators or multicyclones, after which it is converted into magnesium hydroxide ($Mg(OH)_2$) upon addition of water. This slurry of $Mg(OH)_2$ in water is purified *via* sedimentation or filtration and reacted with sulphur dioxide or elemental sulphur to form the cooking acid magnesium bisulphite. The other method comprises stripping columns to remove excess SO_2 from the condensate to avoid disturbance of the bio-organisms in the anaerobic wastewater treatment.

The process focuses on the separation and utilization of hemicellulose and lignin into their derivatives such as lignosulfonates, ethanol, fodder yeast, soda, vanillin, acetic acid and furfural.[31]

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

The main advantage of the sulphite pulping process is the high flexibility of the cooking process. In fact, the entire pH range can be covered by varying the dosage and composition of the chemicals, which allows for the production of a broad range of types and qualities of pulp. The main disadvantage of the sulphite pulping process is the lower strength of the pulp, which makes it a less desirable process in comparison to the Kraft pulping process.[31]

3.3. Organosolv pulping

Organosolv pulping has been investigated for more than 80 years.[32] Pulping studies with organic solvents, *i.e.* with alcohol and water, started already in the 1930s, and by the 1960s a rather wide array of solvents had been identified to be used with or without an acid catalyst.[33] The first patent involving an industrial process for organosolv pulping was filed in 1971.[34] In this patent a process was provided for the pulping of lignocellulosic biomass with a mixture of water and a water miscible volatile organic solvent. Many organosolv pulping papers have been published since. Numerous combinations of solvents, so-called organosolvs, and catalysts are known, and have been applied mainly at the lab scale.

One of the most popular organosolv pulping methods uses 50–100% of methanol in water as solvent. It is mainly applied to the wood species spruce, pine, beech and aspen. The catalysts in this case vary between acids (HCl, H_2SO_4 and methyl sulphuric acid) and metal salts ($CaCl_2$, $Ca(NO_3)_2$, MgCl, $Mg(NO_3)_2$ and $MgSO_4$). The cooking temperatures used are 130–220 °C.[35]

Another often applied option is the use of 40–60% ethanol in water. This organosolv is tested on a broad variation of biomass including spruce, pine, beech, aspen, birch, eucalyptus, red oak, sweet gum, bagasse and rice straw. The catalysts used in this system are acids (HCl, H_2SO_4, and aromatic organic acids), bases (NH_3 and NaOH) and metal salts (Na_2CO_3, $CaCl_2$, $FeCl_3$ and $AlCl_3$). Moreover, $(NH_4)_3PO_4$, anthraquinone and methylanthraquinone can be used. These experiments were conducted at temperatures ranging from 120 to 140 °C.[35]

Many other solvents, including butanol, propanol and glycerol, are described elsewhere.[35,36]

The main advantage of the organosolv process is that sulphur-free lignin is produced that can be further valorised. However, considerable amounts of organic solvents are required for the dissolution of the lignin, resulting in high costs, high regeneration energy consumption, and hazardous and flammable emissions.[35,37]

3.4. Viscose process

The viscose process is used to manufacture regenerated fibres. Dissolving pulp, *i.e.* wood pulp having high cellulose content (>90%), is used as a raw material in this process. The viscose process involves the production of cellulose xanthate, commonly known as Viscose, by the reaction of cellulose with carbon disulphide (CS_2). The cellulose xanthate is then dissolved in aqueous NaOH. The cellulose can be regenerated with sulphuric acid (H_2SO_4), by which CS_2 and hydrogen sulfide (H_2S) are liberated. The amount of highly pure cellulose that can be dissolved with the Viscose process varies from 8 to 12 wt%.[38] However, only 70–75% of the hazardous chemicals needed for the dissolution can be recycled.[7–9,38]

3.5. Lyocell process

The Lyocell process, also known as the NMMO process, is a more recent application for the dissolution of cellulose. Water-free *N*-methylmorpholine-*N*-oxide, NMMO, is used as solvent for the immediate dissolution of cellulose. About 10–15 wt% of cellulose can be dissolved in NMMO. The dissolved cellulose can be recovered by precipitation upon addition of water to the cellulose–NMMO solution. The fibres that are formed share the same name as the process, Lyocell, and are known for their excellent mechanical properties.[38,39]

The Lyocell process is a direct more advantageous dissolution process compared to the Viscose process, because NMMO is biodegradable and can be washed out with water with a recovery rate of about 99.3%.[38,39] However, unwanted by-products, such as *N*-methylmorpholine (NMM) and morpholine, can be produced and thermal runaways can occur.[7–9,38,39]

4. Innovative solvents: ionic liquids *vs.* deep eutectic solvents

Innovative solvents could enhance the biorefinery to a greener and more sustainable industry. For the replacement of conventional solvents and processes, research has been conducted into two categories of innovative solvents. These categories consist of ILs, which can be subdivided into aprotic ionic liquids (AILs) and protic ionic liquids (PILs), and DESs. ILs and DESs offer advantages over conventional solvents due to their properties, which can be tuned by adjusting the chemical structure and composition of their constituents.

ILs are salts composed of an organic cation and an organic/inorganic anion, with melting temperatures below 100 °C and are often liquid at room temperature.[40] The key properties of ILs are negligible volatility, wide liquid range, high conductivity and high thermal stability. ILs are commonly regarded as designer solvents due to the large number of possible cation/anion combinations, which leads to a wide tunability of their physicochemical and toxicological properties.[40] Compared to conventional organic solvents, the unique properties of ILs favour their use in a variety of applications, *e.g.* synthesis and catalysis, electronic devices, thermal fluids and separations.[41–45]

Generally, DESs are formed by mixing a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA), being associated with each other by means of hydrogen bond interactions, upon which a eutectic mixture with a melting temperature much lower than its constituents is formed.[46,47]

DESs share the solvent characteristics of ILs, offering the possibility to tune their physicochemical properties by a suitable selection of their components in terms of molecular structure, chemical nature, ratios and water content.[46–49] Despite still being a relatively young field, DESs already have been used in different applications such as metal electrodeposition,[48] catalysis,[50] drug

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

solubilisation,[51,52] and separation processes.[53–55] Furthermore, a large number of DESs based on natural compounds such as organic acids, amino acids, and sugars were reported.[56,57] Later on, due to the generally high viscosity of these solvents, their properties have been tuned by the addition of water.[58]

It is often claimed that DESs offer advantages over ILs, such as a low price, chemical inertness with water and easy preparation by mixing their constituents at moderate temperature. Furthermore, it is expected that most DESs are biodegradable, biocompatible and non-toxic.[47] Nevertheless, to avoid generalization it should be noted that all these characteristics are derived from the DESs' constituents.

4.1. Preparation of ionic liquids and deep eutectic solvents

The increasing interest in ILs as possible alternatives for the current hazardous organic solvents led to an advancement in their synthetic methods and purification.[40,44] Several methods exist for the preparation of ILs, where the ability to obtain a product of high purity is an important selection criterion.[44] Depending on whether it is an AIL or a PIL, the synthesis is conducted by different procedures, which have been reviewed in the literature.[40,44,59–61]

The most common synthesis of AILs is the quaternization of an amine, phosphine or sulfide with haloalkanes upon the formation of a halide salt. The second step is the anion metathesis reaction by addition of a metal salt, whereby the halide anion is exchanged for the salt anion. In Fig. 4 the most widely used cations and anions of AILs are shown.

A more detailed description of the several synthetic methods to synthesize ILs can be found in the literature.[40,44,62,63]

Despite the fact that the first IL prepared in 1914 was a PIL, AILs have largely dominated the literature. However, research into PILs has recently gained popularity mainly due to their potential for application in fuel cell technologies and biomass fractionation.[8,64] PILs are easily prepared by proton transfer from a Brønsted acid to a Brønsted base, in equimolar proportions. The preparation of a PIL is most commonly an exothermic reaction, so cooling is necessary.

In Fig. 5 some representative cations and anions for PILs are presented.

For PILs it has to be considered that there is a possibility that they are not only composed of ions, but slightly neutral (<1%) due to the dependence on the acid–base equilibrium.[61]

Unlike (A)ILs, DESs are generally easy to prepare. In the most easy and commonly used preparation method, the heating method (mixing under heating), no solvent is needed and no reaction occurs. Consequently, no purification steps are needed, making them promising and economically viable alternatives for conventional organic solvents.[48,65,66]

Other methods reported for the preparation of DESs are the vacuum evaporating, grinding and freeze-drying methods.[47–49,65,67,68] In the evaporating method, the components are dissolved in water, after which the majority of water is evaporated at 50 °C under vacuum. The final mixture is kept inside a desiccator with silica gel until it reaches a constant weight.[68] In the grinding method the two solid components are added to a mortar, after which they are grinded until a clear, homogeneous liquid is formed in the

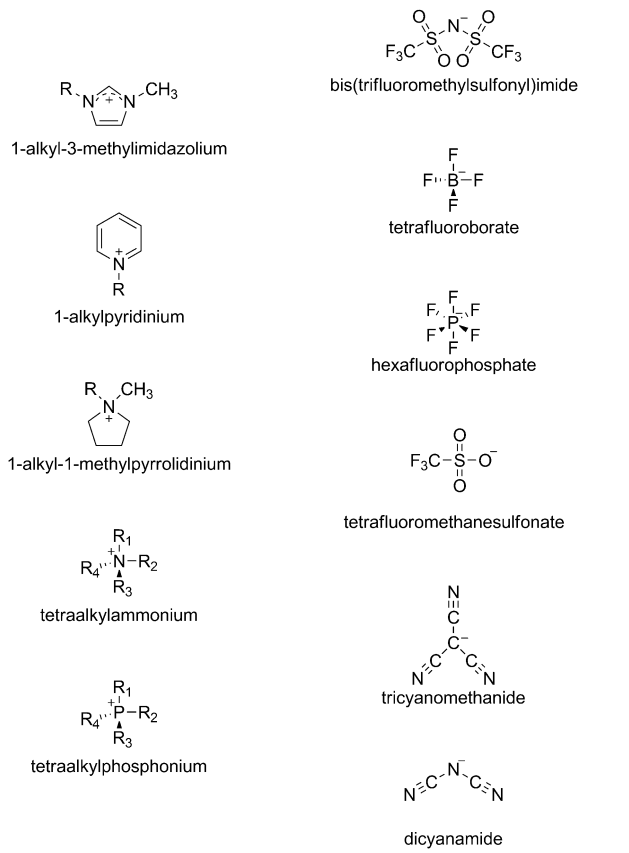


Fig. 4 Commonly used cations and anions for aprotic ionic liquids.

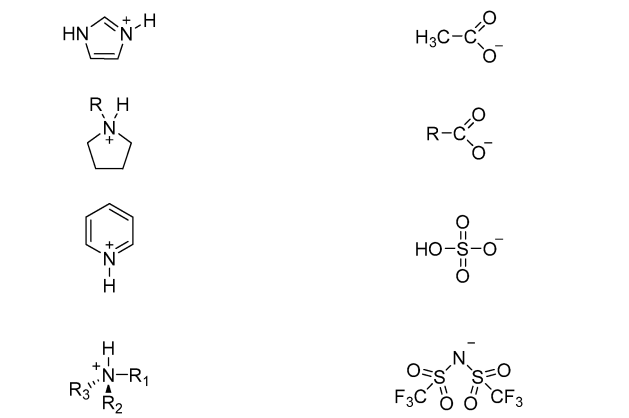


Fig. 5 Commonly used cations and anions for protic ionic liquids.

nitrogen atmosphere of a glovebox.[69] In the freeze-drying method, both the HBD and the HBA are dissolved in approximately 5 wt% of water. These two solutions are mixed, frozen and subsequently freeze-dried to form clear, homogeneous mixtures.[70] Fig. 6 illustrates some of the HBDs and HBAs that have been studied to date.

4.2. Physicochemical properties of ionic liquids and deep eutectic solvents

The main advantage of these innovative solvents is the possibility to tune their physicochemical properties by varying the nature

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

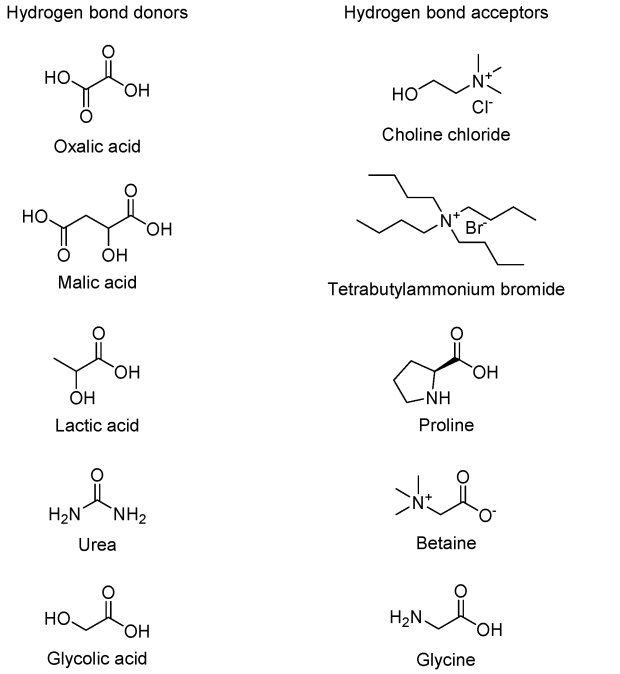


Fig. 6 Structures of some hydrogen bond donors and hydrogen bond acceptors used to form deep eutectic solvents.

(and also ratio in the case of DESs) of the constituents. The most important macroscopic properties to assess the possible use of a given liquid solvent in a particular application are melting temperature, thermal stability, viscosity, density, vapour pressure, toxicity, environmental impact and costs.[61,71,72] These properties for ILs and DESs have been reviewed several times in the literature and are not described here in detail.[40,44,47,49,58–61,71–75] Therefore, only a brief description of thermal phase behaviour, density, viscosity and toxicity is presented.

4.2.1. Thermal phase behaviour. Both AILs and PILs exhibit a wide liquid range, which depends on the lower temperature limit (glass formation or crystallization) and the upper temperature limit (usually the thermal decomposition temperature). For many ILs, the cooling of the liquid phase goes through a metastable state (supercooling), and later to a glass transition, which is detected typically in the region of −60 to −100 °C. In some cases multiple solid–solid transitions have been observed.[76–78]

The upper temperature limit for ILs is usually related with the thermal decomposition temperature, mainly due to their low volatility. Most of the ILs can be kept in the liquid phase up to 400 °C, showing thermal decomposition around 300 to 430 °C, depending most strongly on the choice of the anion.[73]

DESs are usually composed of asymmetric components contributing to a decrease of the lattice energy and consequently lower melting temperatures.[48] The melting temperature of DESs is related to the magnitude of the interactions between the HBD and HBA. DESs are characterized by having a lower melting temperature than the individual components. DESs that are liquid at room temperature are the focus of intense research.[47–49,79]

Like ILs, some room temperature DESs do not crystallize, but undergo a supercooled state or glass transition.[55,68,69] In other cases, their thermal behaviour is quite complex and they can exhibit multiple solid–solid transitions or plastic crystal behaviour.[80,81] The glass transition of DESs is typically detected in the region of −60 to −100 °C, as observed for ILs.[47–49,55,65,69,82] Regarding the thermal stability, most of the reported DESs present a decomposition temperature around 200 °C or below, being relatively less stable than ILs.[47–49,55,68]

4.2.2. Density. The density of a material depends on how the molecules can pack together, which in turn is related to the size and shape of their molecules and intermolecular interactions. The densities of most of the ILs range between 800 and 1600 kg m^{-3}.[83–86] Literature reviews of thermophysical properties have been published, describing the main aspects of the effect of the cation or anion, impurities and methodologies.[40,84–91]

DESs present densities within the same range as that of the ILs, but higher than that of water. Recently, hydrophobic DESs have also been reported in the literature, with densities considerably lower than that found for most of the reported hydrophilic DESs.[80,92,93] Following the trend that was observed for ILs, the densities of DESs decrease with an increase of the alkyl chains of their components.[47,55,69,80,94–97] It was also observed that the densities decrease as the relative ratio of the salt to HBD is increased.[98,99]

4.2.3. Viscosity. Viscosity is an important property, and it describes the internal resistance of a fluid to shear stress. At ambient temperature, ILs present a higher viscosity (around 10 mPa s to 726 mPa s) than typical molecular liquids (around 0.2 mPa s to 10 mPa s).[40,84,100] The viscosity of an IL widely depends on the combination of cation and anion or impurities present, and their effects are well studied in the literature.[40,71,84,101,102] For example, the presence of very low concentrations of chloride drastically increases the viscosity, whereas the presence of water significantly reduces the viscosity.[103–108]

DESs exhibit relatively high viscosities (>100 mPa s at ambient temperature). Depending on the combination of the HBD and HBA viscosities above 10 000 mPa s can even be reached.[109] The viscosity of a DES is mainly affected by the chemical nature and ratio of its components, the temperature and the water content. Recently, the addition of water has been presented as a method to tailor the viscosity of DESs.[58,70,110] However, it should be noted that individual DES components are fully solvated at water contents between 25 and 50 wt%.[58,110] Hence, if the DES' interactions are to be preserved, there are limits to the addition of water to tailor the viscosity. In general, in a viscous liquid, such as an IL or a DES, the decrease of viscosity with increasing temperature is more pronounced than in a molecular liquid.[47,55,69,80,84,87,96,111,112] The higher viscosities are directly related with their intermolecular interactions. The increase of the temperature will considerably decrease their intensity and consequently decrease the viscosity.

4.2.4. Toxicity. ILs and DESs are usually regarded as 'green' solvents, especially due to their low volatility. However, the high solubility in aqueous media represents a risk to the aquatic environment. The stability of ILs makes them poorly decomposable by microorganisms,[113–116] persisting in the environment for some time.[117] The antimicrobial activity towards microorganisms in a wide range of ILs was investigated.[56,118] Most importantly,

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

an increase of toxicity with the increase of the cation alkyl chain length was observed. One of the main advantages of DESs is the promising low toxicity compared to ILs. DESs have usually been considered as non-toxic solvents, due to the use of natural components for their preparation. The toxicity of phosphonium-based DESs was evaluated and it was found that the toxicity depends on the composition, viscosity and ratio of each component.[119] Additionally, the toxicity of the DESs was compared with that of the individual constituents, and it was found that the studied DESs presented either higher or lower toxicity.[119] More information on toxicity of DESs can be found in the literature.[79,117,120]

5. Lignocellulosic fractionation by using ionic liquids and deep eutectic solvents

Currently, cellulose and lignin are two of the most investigated components of lignocellulosic biomass. For example, many papers have been published on the conversion of lignin and cellulose into their building blocks, such as glucose for cellulose and aromatic groups for lignin. However, these investigations can only be conducted after fractionation and processing of the lignocellulosic biomass, which will be discussed in this chapter. The focus will be on the dissolution of cellulose and lignin and the extraction of these constituents from lignocellulosic biomass with ILs and DESs. Specifically, the IL/DES properties that are beneficial for biomass dissolution will be evaluated.

5.1. Cellulose dissolution with ILs

5.1.1. Which ILs can dissolve cellulose? The first publication on the dissolution of cellulose in an IL originates from 2002 when up to 25 wt% of cellulose was dissolved in the IL 1-butyl-3-methylimidazolium chloride [C_4mim][Cl] under microwave heating and up to 10 wt% upon heating to 100 °C.[121] Until now, 2276 papers related to this topic have been published (Scopus March 2015 with the search entries 'ionic liquids' followed by 'cellulose dissolution'). One of the major problems related to the comparison of results in the literature is the many different types of pure cellulose available on the market, *e.g.* Avicel, microcrystalline cellulose (MCC), pre-hydrolysis sulfate pulp, α-cellulose. They all have different characteristics, such as molecular weight and morphology, which are not always completely analysed. Especially an increase in molecular weight will lead to a decrease in dissolution ability and subsequently to the difficulties in comparing results. In addition, comparison is complicated due to the wide variety of dissolution conditions. For instance, temperatures varying between 25 °C and 110 °C have been reported. A broad overview of ILs able to dissolve cellulose is described elsewhere.[7–9]

5.1.2. What is the role of the anion? The influence of the anion was already noticed in the first publication about the dissolution of cellulose with ILs. With the use of $[C_4mim]^+$ as cation, five different anions were used for the dissolution of cellulose *via* microwave heating, *i.e.* $[Cl]^-$, $[Br]^-$, $[SCN]^-$, $[BF_4]^-$ and $[PF_6]^-$. The use of the anions $[BF_4]^-$ and $[PF_6]^-$ led to no dissolution, while the anions $[Br]^-$ and $[SCN]^-$ gave a cellulose dissolution of 5–7 wt%. The use of $[Cl]^-$ as anion introduced cellulose dissolution as much as 25 wt%. These results clearly state the high importance of the anion.[121] In 2008 an investigation was conducted to investigate the influence of phosphonium containing anions, namely 1-buyl-3-methylimidazolium methylphosphonate [C_4mim][(MeO)(H)PO_2], 1-butyl-3-methylimidazolium methyl methylphosphonate [C_4mim][(MeO)(Me)PO_2] and 1-butyl-3-methylimidazolium dimethylphosphate [C_4mim][(MeO)(MeO)PO_2]. It was expected that [C_4mim][(MeO)(Me)PO_2] had the largest cellulose (MCC) dissolution capability since this IL has the highest Kamlet–Taft β parameter. However, it was observed that the IL [C_4mim][(MeO)(MeO)PO_2] had a higher ability to dissolve cellulose at low temperatures, which was contributed to its lower viscosity.[122] The first observation of the screening of ionic liquids for the dissolution of cellulose in 2008 was the fact that hydrophobic ILs were not able to dissolve cellulose. Furthermore, it was observed that dicyanamide as anion could only dissolve up to 1% of cellulose, so it can be considered useless for this application. On the other hand, ILs with chloride are able to dissolve cellulose, although they are not considered as preferable solvents due to their high melting points, their hygroscopic behaviour and their toxic behaviour. ILs composed of formate anions generally show high solvability for cellulose, but they also show low thermal stability caused by decarboxylation at high temperatures. Finally, it was concluded that ILs containing acetate could dissolve substantial amounts of cellulose and were thermally stable.[123] NMR studies into chloride anions showed large $^{35/37}$Cl relaxation parameters, indicating hydrogen bonding interactions between the chloride anion and the hydroxyl groups of cellulose. Changes in the ^{13}C relaxation parameters for the IL [C_2mim][CH_3COO] were also observed, although less pronounced.[124] Next to the more general accepted anions for the dissolution of cellulose, it was shown in 2009 that the ILs composed of phosphonium anions, 1-ethyl-3-methylimidazolium diethyl phosphate ([C_2mim][Et_2PO_4]) and 1,3-dimethylimidazolium dimethyl phosphate ([C_1mim][Me_2PO_4]), were able to dissolve 14 wt% and 10 wt% of cellulose. An increase of the alkyl chain in the anion of these ILs led to a decrease in cellulose dissolution capacity. Molecular dynamics between the [C_2mim][CH_3COO] and glucose as the model compound for cellulose showed that the acetate anion has hydrogen bonding interactions with the three hydroxyl groups of glucose. In a comparison between water and the above mentioned IL it was observed that the hydrogen bonding energy between water and glucose was considerably lower than that between the IL and glucose (5 kcal mol^{-1} *vs.* 14 kcal mol^{-1}).[125] In 2010 a publication focused specifically on the effect of the anion on the dissolution of MCC. A decrease in cellulose solubility was detected over the following range of anions: $[CH_3COO]^- > [HSCH_2COO]^- > [HCOO]^- > [(C_6H_5)COO]^- > [H_2NCH_2COO]^- > [HOCH_2COO]^- > [CH_3CHOHCOO]^- > [N(CN)_2]^-$. It was observed that the introduction of an electron withdrawing group, such as OH, SH, NH_2 or CH_3OH, decreased the solubility of cellulose.[126] Furthermore, they concluded that

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

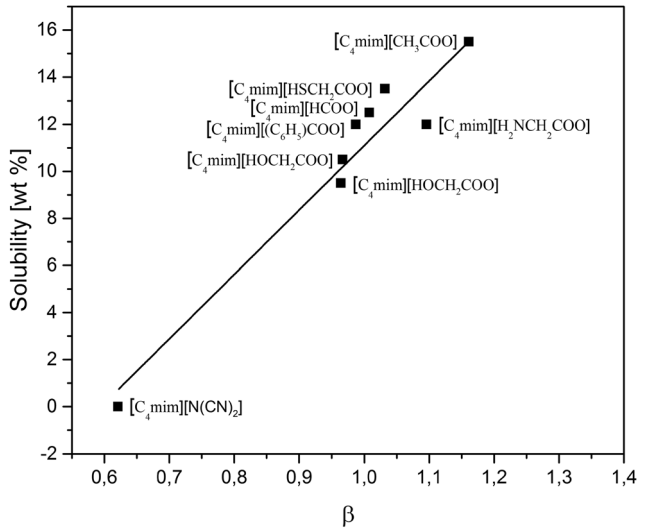


Fig. 7 Proposed linear correlation between the Kamlet–Taft β parameter and the cellulose solubility in the IL[126] (reproduced with the data available in the reference).

hydrogen bond accepting strength played an important role and a linear correlation was proposed between the solubility of cellulose in wt% and the Kamlet–Taft β parameter as shown in Fig. 7.

The Kamlet–Taft parameters were first introduced in 1976 by Kamlet and Taft. The Kamlet–Taft parameters consist of five parameters, *i.e.* the Kamlet–Taft parameter $E_T(30)$, E_T^N, π^*, α, and β.[127] The $E_T(30)$ and E_T^N describe the interactions between the solvent and the dyes used for the measurements and so express the solvent polarity. The Kamlet–Taft parameter π^* defines the polarizability/dipolarity of a solvent, which is the ability of a solvent to stabilize a charge or a dipole by virtue of its dielectric effect. The Kamlet–Taft α parameter describes the hydrogen bond donating strength, also commonly known as hydrogen bond acidity, while the Kamlet–Taft β parameter is used for describing the hydrogen bond accepting strength, also commonly known as the hydrogen bond basicity.[127]

In general, ILs have a high π^*, a moderate α and a high β. Moreover, α depends mainly on the cation, while β depends on the anion. It was found that the Kamlet–Taft β parameter is most important for the dissolution of cellulose. Dissolution of cellulose occurs when the β parameter is higher than approximately 0.8.[8] Furthermore, it can be observed that the solubility is at its maximum with the highest Kamlet–Taft parameter, giving a clear correlation (Fig. 7).

A more recent study showed that the electrostatic part of the interaction between cellulose and the anion $[Cl]^-$ is approximately 10 times stronger than the interaction of cellulose with the cation $[C_4mim]^+$.[128] This has been confirmed by another molecular dynamics investigation.[129]

5.1.3. What is the role of the cation? Tables 1 and 2 present the Avicel cellulose solubility in different ILs with the same anion, but with a variation in cation. These data are used to study the effect of the cation on cellulose solubility. It can be noticed that ILs containing imidazolium, ammonium and morpholium cations can dissolve considerable amounts of cellulose. Previous results indicated that ILs consisting of a pyridinium or phosphonium as cation are also able to dissolve cellulose.[7] However, there is no general consensus on the role of the cation in the ability of the IL to dissolve cellulose. This is partially due to contradictions in the literature.

Table 1 Influence of the cation on the dissolution of cellulose at a temperature of 110 °C[123]

Ionic liquids	Solubility [wt%]
$[C_2mim][CH_3COO]$	15
$[C_8mim][CH_3COO]$	<1
$[AMM110][CH_3COO]$	0.5
$[Me(OEt)_2\text{-Et-Im}][CH_3COO]$	12
$[Me(OEt)_3\text{-Et-Im}][CH_3COO]$	12
$[Me(OEt)_4\text{-Et-Im}][CH_3COO]$	10
$[Me(OEt)_7\text{-Et-Im}][CH_3COO]$	3
$[Me(OEt)_3\text{-MeOEtOMe-Im}][CH_3COO]$	0.5
$[H(OEt)_2\text{-Me-Im}][CH_3COO]$	5
$[H(OEt)_3\text{-Me-Im}][CH_3COO]$	2
$[Me(OPr)_3\text{-Et-Im}][CH_3COO]$	0.5
$[Me(OEt)_3\text{-Bu-Im}][CH_3COO]$	<0.5
$[Me(OEt)_3\text{-Et}_3N][CH_3COO]$	10
$[Me(OEt)_2\text{-Et}_3N][CH_3COO]$	10
$[MM(EtOH)NH][CH_3COO]$	<0.5
$[(MeOEt)_2NH_2][CH_3COO]$	<0.5
$[MM(MeOEt)NH][CH_3COO]$	<0.5
$[M(MeOEt)_2NH][CH_3COO]$	<0.5

Table 2 Influence of the cation on the dissolution of cellulose at 120 °C[130]

Ionic liquids	Solubility [g mol^{-1}]
$[C_4mim][CH_3COO]$	58
$[C_1OC_2mim][CH_3COO]$	56
$[C_2OHmim][CH_3COO]$	34
$[C_4dmim][CH_3COO]$	37
$[phC_1min][CH_3COO]$	34
$[C_2mmor][CH_3COO]$	31
$[C{=}C_2mmor][CH_3COO]$	28
$[C{=}C_2mpip][CH_3COO]$	19
$[C_4mpip][CH_3COO]$	7
$[C_4mpyr][CH_3COO]$	3
$[C_4ebim][CH_3COO]$	<1
$[C_2ebim][CH_3COO]$	<1
$[C_4ebt][CH_3COO]$	<1

^{13}C NMR studies on cellobiose and glucose, in 2006 and 2008, respectively, concluded that the influence of the cation on the dissolution of cellulose is minor, due to only small interactions between the cationic part of the IL and the sugars.[124,131] However, in both 2010 and 2014, ^{1}H NMR and ^{13}C NMR studies on cellobiose and cellulose suggested that the acidic protons, especially the proton on the H2 position of the imidazole ring, formed hydrogen bonding interactions with the oxygen atoms of cellulose.[132,133] Before any further discussion, it should be noted that some results were questioned in previous articles.[134,135] This was because conclusions were mainly based on chemical shift perturbation ($\Delta\delta$) data, which are claimed to be improper indicators, since they only give information about changes in the chemical environment. However, it was shown that replacing the acidic proton on the 2-position of the imidazole ring with a methyl group led to a decrease in cellulose dissolution from

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

59 g mol^{-1} to 37 g mol^{-1} (comparison between [C_4mim][CH_3COO] and [C_4dmim][CH_3COO]).[130] This can be seen as a strong effect, though the question still remains whether this is caused by a lack of hydrogen bonding or as a result of a change in the polarity or hydrogen accepting ability of the IL. Unfortunately, the Kamlet–Taft parameters are not given for the [C_4dmim][CH_3COO] IL, so a good comparison between this IL and the [C_4mim][CH_3COO] is rather difficult. Especially a comparison of the Kamlet–Taft β parameters of these two ILs, the parameter that is considered as the most influential, would give some more data for discussion.

The interaction between IL and cellulose, or alternatively between IL and glucose/cellobiose as the model compound for cellulose, has also been investigated with molecular dynamics studies. Initial molecular modelling on D-glucose solvation in the IL 1,3-dimethylimidazolium chloride ([C_1mim][Cl]) showed that the cation only has weak interaction with glucose.[136] It was considered that the interaction between the cation and the chloride anions, which are hydrogen-bonded to glucose, resulted in regions of cation density around the sugars as a secondary effect. These studies were continued with a 1 : 5 β-glucose : [C_1mim][Cl] ratio to obtain a mixture more representative and comparable to experimental data. In these studies it was observed that some weak hydrogen bonding interactions occurred between the hydrogen on the 2-position of the cation and the oxygen on the 6-position of glucose.[137] It was concluded that the decrease in relaxation rates as observed in NMR studies can be attributed to the presence of interactions between the cation and the glucose. From molecular modelling on [C_4mim][Cl] it was concluded that the cation does not directly interact with cellulose. The prevention of interaction between the IL and glucose was attributed to steric hindrance.[138] Furthermore, investigations were conducted on the interactions between [C_2mim][CH_3COO] and 1–4 linked β-D-glucose oligomers with a polymerization degree of 5, 6, 10 and 20. They concluded that the imidazolium ring contributes to the dissolution of cellulose, since they observed that the conjugated ring structure had van der Waals interactions with sugar rings. In the same publication it was also observed that the cations were in close contact with the polysaccharides through hydrophobic effects.[125] In 2013 it was observed that the cation in this research, [C_4mim]$^+$, stacks on the hydrophobic part of the cellulose surface by means of non-polar interactions.[139]

In contrast to previously discussed studies which showed that the possible interaction between the cation and the cellulose is mainly *via* hydrogen bonding, van der Waals interactions and hydrophobic effects, some investigations indicate that there is covalent bonding. The interactions of both [C_2mim][CH_3COO] and [C_2mim][Cl] with cello-oligomer were investigated and it was observed that during the dissolution of cello-oligomer in the IL [C_2mim][CH_3COO] the C-1 carbon signal of the cello-oligomer in NMR disappeared.[140] This was explained as the result of the formation of a covalent bond between the carbon at the 2-position of the [C_2mim][CH_3COO] and the carbon at the 1-position of the cello-oligomer. Interestingly, this disappearance was not detected in the carbon NMR of the cello-oligomer/[C_2mim][Cl] mixture.[140] The discovery of the covalent bonding was verified by an investigation of the formation of the covalent bonds between ILs, butylmethylimidazolium acetate ([C_4mim][CH_3COO]) and 1-(2-napthylmethyl)-3-methyl-imidazolium acetate ([NapMIM][OAc]), and glucose *via* ^{13}C-isotopic labelling and fluorescence labelling experiments.[141] The results confirmed that the C-2 carbon of the cation of the IL is covalently attached to the reducing end of cellulose. Furthermore, it was pointed out that this reaction can be suppressed by the absence of bases.[141]

There is also a clear effect of the length of the alkyl chain on the cellulose dissolution behaviour. Generally, an increase of alkyl chain length beyond C_4 gives a decrease of cellulose solubility. A solubility screening of 1-alkyl-3-methylimidazolium chlorides was performed with a variation of the alkyl chain length from C_2 to C_{10}. Surprisingly, the authors found a strong effect between an even and an odd alkyl chain where the even alkyl chains were superior in dissolution ability over the odd alkyl chains. For the even chains the C_4 alkyl chain showed the best dissolution behaviour up to 20 wt% (just after C_2), while for the odd chain this was the C_7 alkyl chain, which dissolved only up to 5 wt%.[142] Reports show that no odd/even effect was noticed for a variation in alkyl length with bromide as anion, although this conclusion is questioned, since the maximum solubility was only 2–3%.[143] It was reported that the decrease in solubility over an increase of alkyl length might be caused by an increase of viscosity since the solution becomes so viscous that more dissolution is not possible.[123] The authors exemplify this with the increase of viscosity of the methylimidazolium acetate ILs, where an increase of C_2 to C_4 led to an increase of viscosity of 162 mPa s to 646 mPa s.[144,145] Thus, for longer carbon chains this would lead to extreme viscosities and much lower dissolution rates.

Moreover, functional groups within the alkyl chain groups could contribute to an increase or decrease of the cellulose dissolution ability. The higher dissolution ability of 1-ethyl-(3-(2-(2-methoxyethoxy)ethoxy)ethyl)imidazolium acetate ([Me(OEt)$_3$-Et-Im][CH_3COO]) and 1-ethyl-(3-(2-(2-(2-methoxyethoxy)ethoxy)-ethoxy)ethyl)imidazolium acetate ([Me(OEt)$_4$Et-Im][CH_3COO]) over 1-methyl-3-octylimidazolium acetate ([C_8mim][CH_3COO]) was contributed to the presence of an oxygen atom in the alkyl chain.[123] It was hypothesized that due to the presence of oxygen in the alkyl chain of the cation it could act as a Lewis-base/hydrogen bond acceptor leading to increased hydrogen bonding with cellulose and subsequently more cellulose dissolution.[123] However, Wang *et al.* observed that the solubility of cellulose diminished when the –C_4H_9 alkyl chain was replaced with –$CH_2CH_2OCH_3$ and CH_2CH_2OH. They attributed this decline to the electron donating behaviour of the –OCH_3 and –OH.[130] Thus, the presence of oxygen can have a positive influence and a negative influence probably based on the nature of oxygen which can behave as an electron donating group on aromatic groups and as an electron withdrawing group in alkyl chains. Zhang *et al.* concluded from molecular dynamics that electron withdrawing groups indeed enhanced the interaction with cellulose and led to an increase in cellulose solubility. They claimed that this phenomenon originated from an increase of electronegativity of the cation.[132] Furthermore, there is a possibility that the hydrogen of the hydroxyl group undergoes intermolecular hydrogen bonding with the anion.[146] This could lead to competition

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

with the hydrogen bonding of the anion with cellulose. The precise influence of the hydroxyl group is not clear yet, but it was suggested that longer alkyl chains containing hydroxyl groups are more flexible than the shorter alkyl groups giving a better possibility for intermolecular hydrogen bonding.[7]

5.1.4. What is the effect of solvent addition? Many publications have investigated the influence of adding a solvent to the ILs, where both co-solvency and anti-solvency have been investigated. Co-solvents increase the cellulose dissolution ability, whereas anti-solvents hamper it. In 2011 a paper about the dissolution of cellulose in organic electrolyte solutions, which are combinations of molecular solvents with a small to medium molar fraction of IL, was published. A combination of 5 g of 1,3-dimethyl-2-imidazolidinone (DMI) with 5 g of the IL [C_4mim][Cl] was able to dissolve 10 wt% of cellulose (Avicel) after 3 min at 100 °C, whereas the dissolution of cellulose in the pure IL [C_4mim][Cl] normally takes more than 10 h.[147] Moreover, when [C_4mim][Cl] was replaced with [C_2mim][CH_3COO], instantaneous dissolution of cellulose occurred with a mole fraction of 0.18 IL. In the same publication DMI was replaced with 14 other molecular solvents. The most striking results, next to that of DMI, were obtained for the combination of the molecular solvents *N*,*N*-dimethylformamide (DMF), *N*-methyl-2-pyrrolidone (NMP) and dimethylsulfoxide (DMSO). In these solvents, respectively, mole fractions of 0.10, 0.16, and 0.08 IL were needed for the direct dissolution of cellulose. In another publication the main focus was on the prediction of suitable co-solvents with the Kamlet–Taft parameters. It was concluded that molecular co-solvents should be rather polar with $E_T^N > 0.3$ and $\pi^* > 0.8$, have a low HBD ability with $\alpha < 0.5$ and a strong HBA ability with $\beta \geq 0.4$.[148] In 2013 it was speculated that adding an aprotic polar solvent, such as DMSO, to an IL would increase the solvation of the cation, leading to more 'free' anions. These anions would then be more accessible for hydrogen bonding interactions with cellulose, increasing its dissolution.[149] They also investigated the dissolution of MCC, with a degree of polymerization (DP) of 229, in pure [C_4mim][CH_3COO], DMSO and a combination of the two. Surprisingly, at 25 °C no dissolution in the pure IL and the pure molecular solvent occurred, while 15% of MCC was dissolved in the mixture. The molar ratio 2.54 : 1 is the optimal ratio for DMSO : [C_4mim][CH_3COO]. Also cellulose samples with a higher DP were dissolved, *i.e.* absorbent cotton (DP = 1586) and native cotton (DP = 2179). At 35 °C, 6.0% of absorbent cotton and 3.5% of native cotton were dissolved, while the dissolution levels were 12.5% and 11.0% at 85 °C. Moreover, they investigated the co-solvents DMF and dimethylacetamide (DMA), also in a 2.54 : 1 ratio, and found that the DMF/[C_4mim][CH_3COO] solvent could dissolve 12.5% and DMA/[C_4mim][CH_3COO] 5.5% of cellulose at 25 °C. The differences in dissolution were contributed to the dipole moment of the aprotic polar solvent, DMSO (3.96 D) > DMF (3.86 D) > DMA (3.81 D), which allows for a higher solvation. A recent publication hypothesized that a pre-treatment of cellulose with DMF would swell the cellulose, leading to increased distances between cellulose strands. This subsequently allows for a quicker interaction between cellulose and IL.[150] This hypothesis was confirmed by NMR studies, in which the interactions between the IL [C_4mim][CH_3COO] and aprotic organic solvents were investigated.[151] Their investigation showed that ethyl acetate facilitated the most enhanced interaction between the IL and cellulose (Alicell-Super). Moreover, it was shown that the protic formamide had the strongest swelling effect, but also introduced competition between the IL and cellulose. DMSO in combination with a crown ether (18-crown-6) and the IL tetrabutylammonium acetate was able to dissolve 8% cellulose (DP_w = 830) within 5 min and 12% in 30 min, both at 40 °C.[152] Moreover, it was claimed that cellulose was molecularly dissolved, *i.e.* no degradation occurred, with a combination of the co-solvents chloroform, DCM, DMF, acetonitrile and propylene carbonate with the IL [C_2mim][CH_3COO] up to co-solvent/IL molar ratios of 10 : 1.[153] They made this conclusion after analysis with static light scattering (SLS) and small angle neutron scattering (SANS), which showed no molecular aggregation.[153] It was found that a novel method, in which the co-solvent was rapidly evaporated, ensured high solubility up to 27 wt% in pure [C_2mim][CH_3COO].[153] Finally, it was concluded that an almost instantaneous dissolution occurred in the solvent mixture with the highest electrical conductivity.[153] An investigation, both theoretically and experimentally, concluded that DMSO improves the dissolution of cellulose in [C_4mim][CH_3COO] + DMSO mixtures due to an enhancement of mass transfer caused by the decrease in viscosity.[154] Furthermore, it was concluded that DMSO had no interactions with the cation and the anion. This was confirmed by another investigation solely based on molecular dynamics.[155] Since 2013 more molecular simulations have been conducted into the dissolution of cellulose with ILs and molecular co-solvents. It was shown that the interaction between cellulose and the anion $[Cl]^-$ increases upon addition of DMSO, while for water the contrary happens due to the hydrogen bonding between the anion and water.[128]

The addition of anti-solvents has also been investigated, which can lead to the precipitation of cellulose and consequently to its regeneration. Arguably, the most investigated anti-solvent is water, which is for two reasons. Firstly, ILs often contain hygroscopic parts taking up water from air. Secondly, most biomass comprises a low percentage of water. Thus, it is of major importance to gain knowledge of the influence of water. The first publication in the field of cellulose dissolution with ILs showed that more than 1 wt% of water hampered the cellulose dissolution. The authors explained this by competitive hydrogen bonding between the water molecules and cellulose.[121] The authors also indicated that for this reason water is an excellent anti-solvent, inducing the precipitation of cellulose. The fact that water decreases the dissolution of cellulose was acknowledged by multiple publications, although the amount of water at which turbidity and/or precipitation occurred varied.[121,143,148,149,156–158] A publication in 2009 investigated the influence of water and cellulose solubility in the IL [C_4mim][Cl].[159] The more the amount of water added, the lower the solubility of cellulose in [C_4mim][Cl].[159] Water is a protic solvent with a high Kamlet–Taft α parameter (1.17) and a moderate Kamlet–Taft β parameter (0.47). The high Kamlet–Taft α parameter could indicate that it interacts with the hydrogen bond accepting anions.[148] Determination of the

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Kamlet–Taft parameters of ILs containing 1 wt% of water showed a decrease in hydrogen accepting ability, owing to a lower Kamlet–Taft β parameter, which might indicate that this is caused by hydrogen bonding interactions of the anion with water.[157] Furthermore, it was found that the water uptake was dependent on the anion, decreasing in the following order: $[CH_3COO]^- > [Et_2PO_4]^- > [(CN)_2N]^- > [CF_3SO_3S]^- > [BF_4]^- > [PF_6]^-$.[143] Remarkably, tetrabutylphosphonium hydroxide (TBPH) and tetrabutylammonium hydroxide (TBAH) were able to dissolve 20 wt% of MCC with a staggering 40 wt% of water in respectively 5 and 6 min at 25 °C. However, it should be mentioned that it is thought that water is essential for the stabilization of these ILs.

More recently, it was concluded that the water content had a small influence on the recovery in comparison to the temperature. At 90 °C it was found that both [C_4mim][Cl] and [C_4mim][Cl] with water behave as non-derivatizing solvents, while at 120 °C they behave as derivatizing solvents.[160] Furthermore, the effect of different amounts of water (0%, 20%, 50%, 80%) on the regeneration of cellulose was investigated at 40 °C. The effect of temperature was further explored at 60 °C and 80 °C with 50 wt% of water.[129] It was concluded that with an increase of water content the cellulose regenerates due to aggregation of cellulose chains in the mixture of the IL [C_4mim][CH_3COO] and cellulose. The increase of water led to a decrease of hydrogen bonding between the acetate group of the IL and cellulose. Consequently, the number of hydrogen bond interactions between the cellulose chains increased. The influence of temperature showed that at higher temperatures the cellulose–cellulose interactions are stronger, which has a positive effect on the regeneration.

Next to water, different anti-solvents have been reported. In general, it is believed that protic solvents are excellent anti-solvents. Examples are ethanol, methanol, 2-propanol, acetone and acetonitrile. In terms of mechanism, it is thought that they work the same way as water. Thus, hydrogen bonding will occur with the anion of the IL or with the cellulose. An investigation into the regeneration with acetone, ethanol and water showed that the strength of cellulose with $[CH_3COO]^-$ decreases from acetone > ethanol > water.[161] Furthermore, there is an increase of hydrogen bonding interactions between cellulose and $[CH_3COO]^-$ over water > ethanol > acetone.[161]

5.1.5. Protic ionic liquids. The main difference between AILs and PILs is the cationic part, which is aprotic for AILs and protic for the PILs. So far, the dissolution of cellulose in AILs was described. In this section the dissolution of cellulose in PILs is presented (see Table 3). The first research into the dissolution of cellulose with PILs originates from 2008 when ILs were presented that could both dissolve carbohydrates and were compatible with enzymes. In that work, the authors investigated 32 AILs and 4 PILs. The PILs used in this investigation were acetic acid in combination with *N*,*N*-dimethylethanolammonium acetate ([MM(EtOH)NH][CH_3COO]), bis(2-methoxyethyl)ammonium acetate ([$(MeOEt)_2NH_2$][CH_3COO]), *N*,*N*-dimethyl-2-methoxyethylammonium acetate [MM(MeOEt)NH][CH_3COO] and *N*-methyl-bis(2-methoxyethyl)ammonium acetate [$M(MeOEt)_2NH$]-[CH_3COO].[123] All four PILs showed cellulose (Avicel PH-101) dissolution ability lower than 0.5 wt% at a temperature of 110 °C. In 2010 an investigation regarding the dissolution of cellulose in alkanolammonium-based PILs was published.[162] The alkanolammonium PILs were prepared by combining an organic acid with amines. The organic acids used were formic acid, acetic acid, malonic acid and citric acid. The amines used were 3-hydroxypropylamine, diallylamine, 2-hydroxyethylamine, bis(2-hydroxyoethyl)amine (diethanolamine) and tris(2-hydroxyethyl)amine (triethanolamine). For the PILs that were formed, no cellulose dissolution was observed (less than 0.1 wt%).

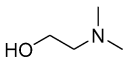
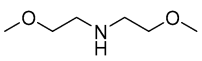
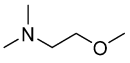
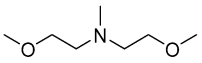
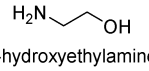
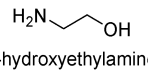
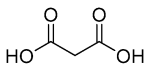
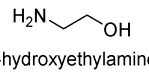
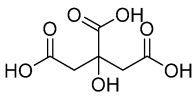
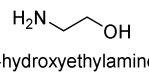
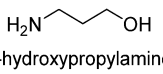
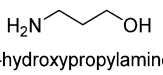
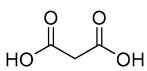
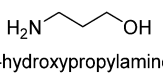
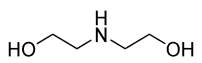
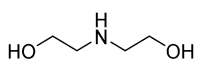
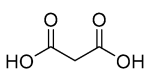
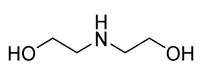
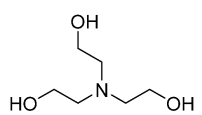
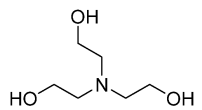
Only in 2011 the first publication involving successful (>5 wt%) cellulose dissolution in PILs was published.[163] Combinations of the superbase 1,1,3,3-tetramethylguanidine (TMG) with the carboxylic acids formic acid, acetic acid, propionic acid, butanoic acid, valeric acid, trifluoroacetic acid and triflic acid were presented. The combination of these acids with the super base led to [TMGH][CO_2H], [TMGH][OAc], [TMGH][CO_2Et], [TMGH][CO_2nPr], [TMGH][CO_2nBu], [TMGH][CO_2nAm], [TMGH][CO_2CF_3], [TMGH]-[OTf] respectively. Furthermore, 1,1,3,3-tetramethylguanidinium bistriflimide ([TMGH][NTf_2]) was prepared and tested for the dissolution of cellulose. The results showed that cellulose could be dissolved with [TMGH][CO_2H], [TMGH][OAc], [TMGH][CO_2Et] and [TMGH][CO_2nPr]. The most rapid dissolution occurred within 10 min at 100 °C with the anions acetate and propionate. Dissolution was also observed with PILs composed of formic acid and butanoic acid as anions, where formic acid as anion has a faster dissolution ability than butanoic acid.

The Kamlet–Taft parameters of two of the above mentioned PILs, [TMGH][OAc] and [TMGH][CO_2Et], were investigated as a function of water content.[157] It was found that the β parameter decreased, as also within the research in AILs, upon addition of water, leading to lower cellulose dissolution. Furthermore, the polarizability/dipolarity (π^*) increased upon water addition, while the Kamlet–Taft α parameter and the $E_T(30)$ parameter remained constant. It was concluded that only the Kamlet–Taft β strength played an important role in the lower solubility of cellulose, while this was not enough for the regeneration. For this reason the authors introduced a second criterion for both the dissolution and the regeneration of cellulose, which they generalized for both AILs and PILs. The criteria for cellulose dissolution that they introduced were $0.35 < \beta - \alpha < 0.9$ and $0.80 < \beta < 1.20$.

In 2013, 12 bases were used for the formation of PILs in combination with acetic acid and propionic acid.[164] They were tested for cellulose dissolution, of which 14 were successful. The reported PILs able to dissolve cellulose are the two above mentioned acids in combination with the 7 bases 1-ethyl-3-methylimidazole-2-ylidene carbene (emim-y), *N*,*N*,*N*,*N*,*N*,*N*-hexamethylphosphorimidetriamide (HMPI), 7-methyl-1,5,7-triazabicyclo[4.4.0]dec-5-ene (MTBD), 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), 1,5-diazabicyclo[4.3.0]non-5-ene (DBN), 1,2-dimethyl-1,4,5,6-tetrahydropyrimidine (DMP) and 1,1,3,3-tetramethylguanidine (TMG). All these PILs were able to dissolve 5 wt% cellulose in the form of pre-hydrolysed Kraft (PHK) pulp at 100 °C. [TMGH][CO_2Et], [DBUH][CO_2Et], [DBNH][CO_2Et] and the AIL [C_2mim][CH_3COO] were further investigated to find the

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

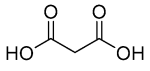
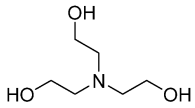
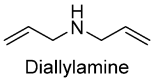
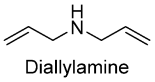
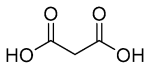
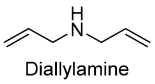
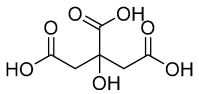
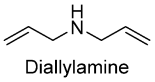
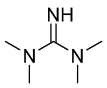
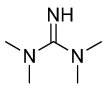
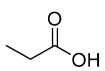
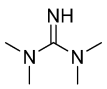
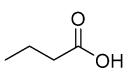
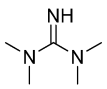
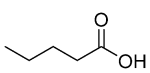
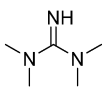
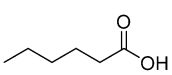
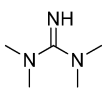
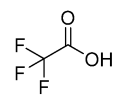
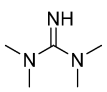
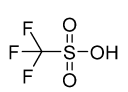
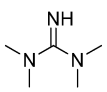
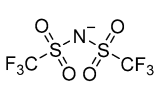
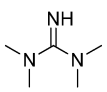
Table 3 Protic ionic liquids for the dissolution of cellulose

Acids	Base	Source of cellulose	T (°C)	t [h]	Sol. [wt%]	Ref‘.
 Acetic acid	 2-Dimethylaminoethanol	MCC (DP 225)	110	1	<0.5	123
 Acetic acid	 bis(2-methoxyethyl)amine	MCC (DP 225)	110	1	<0.5	123
 Acetic acid	 2-methoxy-*N*,*N*-dimethylethan-1-amine	MCC (DP 225)	110	1	<0.5	123
 Acetic acid	 2-methoxy-*N*-(2-methoxyethyl)-*N*-methylethan-1-amine	MCC (DP 225)	110	1	<0.5	123
 Formic acid	 2-hydroxyethylamine	MCC (DP 163)	80	24	<0.1	162
 Acetic acid	 2-hydroxyethylamine	MCC (DP 163)	80	24	<0.1	162
 Malonic acid	 2-hydroxyethylamine	MCC (DP 163)	80	24	<0.1	162
 Citric acid	 2-hydroxyethylamine	MCC (DP 163)	80	24	<0.1	162
 Formic acid	 3-hydroxypropylamine	MCC (DP 163)	80	24	<0.1	162
 Acetic acid	 3-hydroxypropylamine	MCC (DP 163)	80	24	<0.1	162
 Malonic acid	 3-hydroxypropylamine	MCC (DP 163)	80	24	<0.1	162
 Formic acid	 Bis(2-hydroxyethyl)amine	MCC (DP 163)	80	24	<0.1	162
 Acetic acid	 Bis(2-hydroxyethyl)amine	MCC (DP 163)	80	24	<0.1	162
 Malonic acid	 Bis(2-hydroxyethyl)amine	MCC (DP 163)	80	24	<0.1	162
 Formic acid	 Tris(2-hydroxyethyl)amine	MCC (DP 163)	80	24	<0.1	162
 Acetic acid	 Tris(2-hydroxyethyl)amine	MCC (DP 163)	80	24	<0.1	162

This journal is © the Owner Societies 2017

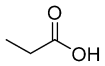
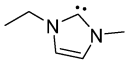
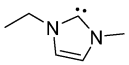
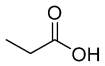
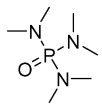
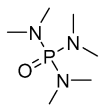
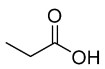
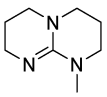
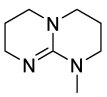
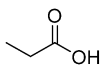
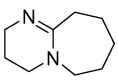
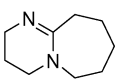
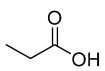
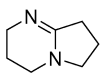
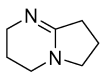
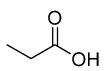
Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Table 3 (*continued*)

Acids	Base	Source of cellulose	T (°C)	t [h]	Sol. [wt%]	Ref‘.
 Malonic acid	 Tris(2-hydroxyethyl)amine	MCC (DP 163)	80	24	<0.1	162
 Formic acid	 Diallylamine	MCC (DP 163)	80	24	<0.1	162
 Acetic acid	 Diallylamine	MCC (DP 163)	80	24	<0.1	162
 Malonic acid	 Diallylamine	MCC (DP 163)	80	24	<0.1	162
 Citric acid	 Diallylamine	MCC (DP 163)	80	24	<0.1	162
 Formic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	5	163
 Acetic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100 105	10 min 20	5 10	163
 Propionic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	10 min	5	163
 Butanoic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	5	163
 Valeric acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	NO	163
 Hexanoic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	NO	163
 Trifluoroacetic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	NO	163
 Triflic acid	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	NO	163
 bis(trifluoromethylsulfonyl)imide	 1,1,3,3-tetramethylguanidine	MCC (DP 381)	100	18	NO	163[a]

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

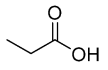
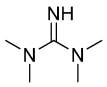
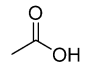
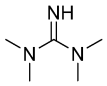
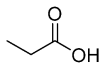
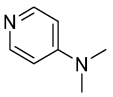
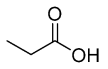
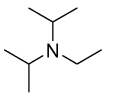
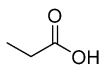
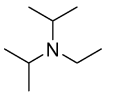
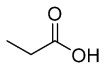
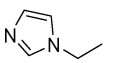
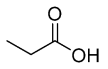
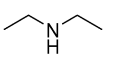
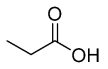
Table 3 (*continued*)

Acids	Base	Source of cellulose	T (°C)	t [h]	Sol. [wt%]	Ref‘.
 Propionic acid	 1-ethyl-3-methylimidazol -2-ylidene carbene	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Acetic acid	 1-ethyl-3-methylimidazol -2-ylidene carbene	Eucalyptus PHK-dissolving pulp	90	N.G.	18	164
 Propionic acid	 Hexamethylphosphoramide	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Acetic acid	 Hexamethylphosphoramide	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Propionic acid	 7-methyl-1,5,7-triazabicyclo[4.4.0]dec-5-ene	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Acetic acid	 7-methyl-1,5,7-triazabicyclo[4.4.0]dec-5-ene	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Propionic acid	 1,8-diazabicyclo[5.4.0]undec-7-ene	Eucalyptus PHK-dissolving pulp	80	N.G.	16	164
 Acetic acid	 1,8-diazabicyclo[5.4.0]undec-7-ene	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Propionic acid	 1,5-diazabicyclo [4.3.0]non-5-ene	Eucalyptus PHK-dissolving pulp	80	N.G.	16	164
 Acetic acid	 1,5-diazabicyclo [4.3.0]non-5-ene	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Propionic acid	 1,2-dimethyl-1,4,5,6- tetrahydropyrimidine	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Acetic acid	 1,2-dimethyl-1,4,5,6- tetrahydropyrimidine	Eucalyptus PHK-dissolving pulp	80	18	5	164

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Table 3 (*continued*)

Acids	Base	Source of cellulose	T (°C)	t [h]	Sol. [wt%]	Ref‘.
 Propionic acid	 1,1,3,3-tetramethylguanidine	Eucalyptus PHK-dissolving pulp	80	N.G.	15	164
 Acetic acid	 1,1,3,3-tetramethylguanidine	Eucalyptus PHK-dissolving pulp	80	18	5	164
 Propionic acid	 4-(dimethylamino)-pyridine	Eucalyptus PHK-dissolving pulp	80	18	NO	164
 Propionic acid	 N,N-di-iso-propylethylamine	Eucalyptus PHK-dissolving pulp	80	18	NO	164
 Propionic acid	 N,N-di-iso-propylethylamine	Eucalyptus PHK-dissolving pulp	80	18	NO	164
 Propionic acid	 1-ethylimidazole	Eucalyptus PHK-dissolving pulp	80	18	NO	164
 Propionic acid	 N,N-diethylamine	Eucalyptus PHK-dissolving pulp	80	18	NO	164
 Propionic acid	 Pyridine	Eucalyptus PHK-dissolving pulp	80	18	NO	164

[a] Acid is already depicted in deprotonated form due to multi-step reaction.

maximum loading of cellulose still dissolvable, which were 15, 16, 16 and 18 wt% respectively.

5.2. Cellulose dissolution with deep eutectic solvents

The research into the dissolution of cellulose with DESs started in 2012 when 26 DESs were tested for the dissolution of lignin, starch and cellulose.[65] Despite the fact that up to 14.9 wt% of lignin could be dissolved with the tested DESs, only minor amounts of cellulose could be dissolved. The lowest solubility, 0.11 wt%, was observed with the DES consisting of malic acid as hydrogen bond donor (HBD) and alanine as hydrogen bond acceptor (HBA) in a 1 : 1 ratio. The DESs comprised of malic acid : glycine and oxalic acid (anhydrous) : proline, both in a 1 : 1 ratio, dissolved 0.14 wt% and 0.15 wt% of cellulose, respectively. Slightly higher amounts of cellulose, 0.24 wt% and 0.25 wt%, were dissolved with different ratios of malic acid : proline (1 : 2 ratio) and oxalic acid (anhydrous) : histidine (9 : 1 ratio). The highest solubility was achieved with the DES composed of malic acid and proline in a 1 : 3 ratio, which was 0.78 wt%. An interesting fact was the increase of cellulose dissolvability over an increase of the ratio of proline in the malic acid : proline DES. Especially the question how the interactions vary over a change in ratio from 1 : 2 to 1 : 3 is very interesting, since this ratio change led to a tripling of cellulose dissolution. Furthermore, it should be mentioned that various temperatures were used for the dissolution, which influenced the amount of cellulose that could be dissolved. The dissolution experiments of the DESs oxalic acid (anhydrous) : proline (1 : 1 ratio) and oxalic acid (anhydrous) : histidine were conducted at 60 °C, while the experiments for the DESs malic acid : alanine (1 : 1 ratio), malic acid : glycine (1 : 1 ratio), malic acid : proline (both the 1 : 2 and the 1 : 3 ratio) were performed at 100 °C.

In 2013, a report mainly discussing chitin dissolution showed 6 DESs that were also able to dissolve cellulose.[165] The 6 DESs showing cellulose dissolvability were urea : choline chloride (2 : 1 ratio), urea : choline bromide (2 : 1 ratio), urea : chlorocholine chloride (2 : 1 ratio), urea : betaine hydrochloride (4 : 1 ratio), glycerol : choline chloride (2 : 1 ratio) and thiourea : choline chloride (2 : 1). Different methods were used as dissolution techniques varying between heating at 100 °C for 10 h, ultrasonication and heating at 80 °C for 1 h and microwave irradiation at 80 °C for 2 h. The results showed that the

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

dissolution of cellulose in the DESs urea : betaine hydrochloride and glycerol : choline chloride was relatively low in comparison to that with the 4 other DESs with maxima of 2.5 and 3.5 wt%. The DESs constituted of urea with choline chloride and urea with betaine hydrochloride had cellulose dissolution abilities in the range of 3.0–5.0 wt% and 5.0–6.0 wt% respectively. Remarkably, the cellulose dissolution in the DES constituted of urea and choline chloride varied between 1.5 wt% and 8.0 wt%. Especially the low cellulose solubility with the microwave irradiation was surprising. The DES composed of thiourea and choline chloride dissolved up to 10 wt% of cellulose with heating at 100 °C, while ultrasonication and heating at 80 °C still led to a dissolution of 6.0 wt%. However, we were not able to reproduce the high dissolution of cellulose in the DES consisting of urea and choline chloride in a 2 to 1 molar ratio. In 2014, a publication showed three DESs with cellulose dissolvability of between 1.03 and 2.83 wt%.[166] The three DESs that were used in this investigation were urea : acetamide (1 : 2 ratio), caprolactam : acetamide (1 : 1 ratio) and urea : caprolactam (1 : 3 ratio). They showed, in the order as depicted above, a solubility of 1.03, 1.79 and 2.83 wt%. However, the temperature applied is not completely clear. In the experimental part it is mentioned that a dissolution temperature of 120 °C is used, while in Table 2 it is said that 50 °C is used. Furthermore, two of the reported eutectic mixtures have melting points higher than room temperature. Thus, it is debatable to define these two eutectic mixtures as DESs. An overview of cellulose solubility in different DESs is presented in Table 4.

5.3. Lignin dissolution with ionic liquids

A considerable amount of research has been performed on finding efficient and non-destructive methods to separate lignin from lignocellulosic biomass. A promising method is its extraction from lignocellulosic biomass with innovative solvents such as ILs. The interest in ILs for the fractionation of lignocellulosic biomass is mainly based on the tunability of their physical properties by suitable selection of cations/anions, their low volatility and their high thermal stability.[57]

The solubility of lignin is most commonly tested using alkaline, Kraft or organosolv lignin. Despite the fact that native lignin differs significantly from lignin obtained *via* commercial methods, it gives valuable information. However, it does not guarantee that an IL with high commercial lignin solubility will also achieve high lignin extractability from lignocellulosic biomass.[8]

5.3.1. Which ionic liquids can dissolve lignin? A number of ILs, both AILs and PILs, have been shown to dissolve lignin to a certain extent (Table 5).[167–170] There is evidence, experimentally and theoretically, that both the cation and anion play an important role in the dissolution of lignin and lignocellulose. It was found that increased *p*-conjugation in the anions or cations of ILs enhances lignin solubility.[171]

5.3.2. What is the role of the anion? Generally, it is assumed that lignin is only soluble in polar ILs with mid-range basic anions.[169,172] An investigation into the effect of the anion showed that lignin was insoluble in the combination of $[Emim]^+$ as cation with weakly coordinating anions such as $[BF_4]^-$, $[PF_6]^-$, and $[(SO_2CF_3)_2N]^-$.[173]

The highly coordinating anions $[O_2P(OCH_2CH_3)_2]$, $[MeSO_4]^-$, $[MeCO_2]^-$, and $[CNS]^-$, as well as the moderately coordinating anion triflate CF_3SO_2, all showed a solubility of approximately 50 wt%. The halide anions with high hydrogen bonding capacity, *i.e.* chloride and bromide, showed a moderate lignin solubility of 10–15 wt%.[168,169]

This observation is backed up by DFT calculations, which show that $[C_1mim][PF_6]$ is 6 kcal mol^{-1} more weakly complexed with a model lignin compound than $[C_1mim][Cl]$.[171] However, a more recent study proposed that the influence of the anion on the degree of dissolution is negligible beyond a critical point.[173]

5.3.3. What is the role of the cation? Also, the cation was found to have an influence on lignin solubility.[169] The highest lignin solubility was found in aromatic cations (*e.g.* imidazolium). A significant reduction in lignin solubility was observed upon methylation of the C2 carbon on the imidazolium ring (*e.g.* $[Bm_2im][CF_3SO_3]$ *vs.* $[C_4mim][CF_3SO_3]$).[173] Furthermore, the IL without any aromatic groups, $[N_{4448}][CF_3SO_3]$, showed evidently the lowest solubility.[173] Replacing the $[C_4mim]^+$ in the IL $[C_4mim][Cl]$ with 1-benzyl-3-methylimidazolium ($[Benzylmim]^+$) improved the dissolution of lignocellulose.

A theoretical study concluded that hydrogen bonding and pi-stacking interactions between lignin and the cation of the IL play an important role in the dissolution of lignin and lignocellulose in ILs.[171] This is consistent with the results that imidazolium cations with *p*-conjugated substituents, such as benzyl and allyl,[174,175] yield more successful lignocellulose solubility. A theoretical model found that imidazolium cations are particularly suitable for lignin solubilisation, as the cation can favourably interact with the lignin phenyl rings *via* the aromatic rings.[171] A high-throughput screening found that the best IL solvent for lignin comprised a combination of $[Cl]^-$ anions and *p*-conjugated $[Amim]^+$ cations.[175]

5.3.4. What is the effect of the polarity of the ionic liquid? Lignin dissolution capacities of ILs can be predicted by comparing their Hildebrand solubility parameters (δH), as a measure of their polarities, with those of lignin. Maximum solubility is observed when the δH values of the biopolymer and solvent are identical, since the solubility of two materials is facilitated when their intermolecular attractive forces are similar.[176] Thus, the high solubility of lignin in $[C_4mim][CF_3SO_3]$ is not surprising given the similarity of the δH values for both the IL and lignin (24.9 and 24.6, respectively). Moreover, the δH values for $[C_4mim][PF_6]$ and $[C_4mim][BF_4]$ are 30.2 and 31.6, respectively,[177] which are sufficiently different from that of lignin, and therefore do not facilitate high lignin solubility.[169]

Another measure to correlate the lignin solubility is the Kamlet–Taft β parameter for hydrogen bond basicity. Mid to high range basic ILs seem to be better solvents for lignin than low hydrogen-bond basic ILs, although hydrogen-bond basicity does not need to be as high as for cellulose (>1) (see Fig. 8).[127]

5.4. Lignin extraction from lignocellulosic biomass with ionic liquids

Lignin extracted with commercial methods is considerably different from its native form. Although solubility tests give

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Table 4 Deep eutectic solvents for the dissolution of cellulose

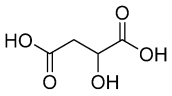
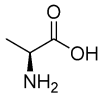
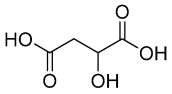
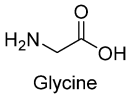
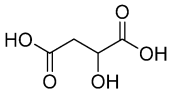
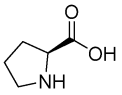
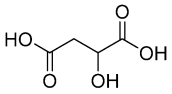
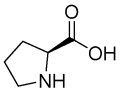
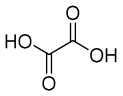
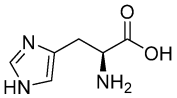
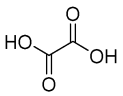
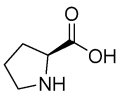
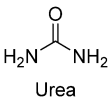
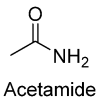
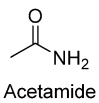
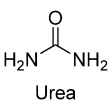
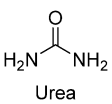
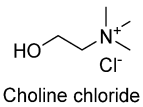
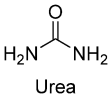
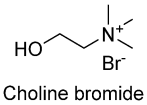
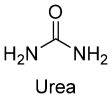
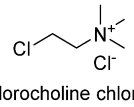
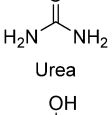
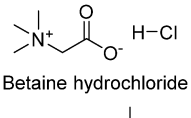
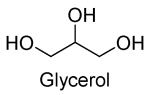
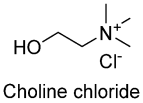
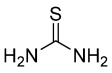
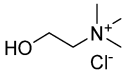
HBD	HBA	Ratio	Source of cellulose?	T [°C]	t [h]	Sol. [wt%]	Ref.
 Malic acid	 Alanine	1 : 1	Cellulose (90%)	100	24	0.11	65
 Malic acid	 Glycine	1 : 1	Cellulose (90%)	100	24	0.14	65
 Malic acid	 Proline	1 : 2	Cellulose (90%)	100	24	0.24	65
 Malic acid	 Proline	1 : 3	Cellulose (90%)	100	24	0.78	65
 Oxalic acid	 Histidine	9 : 1	Cellulose (90%)	60	24	0.25	65
 Oxalic acid	 Proline	1 : 1	Cellulose (90%)	60	24	0.15	65
 Urea	 Acetamide	1 : 2	Cotton–ramie pulp (DP = 517)	120	NA[c]	1.03	166
 Caprolactam	 Acetamide	1 : 1	Cotton–ramie pulp (DP = 517)	120	NA[c]	1.79	166
 Urea	 Caprolactam	1 : 3	Cotton–ramie pulp (DP = 517)	120	NA[c]	2.83	166
 Urea	 Choline chloride	2 : 1	MCC (MW = 3.12 × 10^5 Da)	100 80[a] 100[b]	10 1 2	8.0 6.0 1.5	165
 Urea	 Choline bromide	2 : 1	MCC (MW = 3.12 × 10^5 Da)	100 80[a] 100[b]	10 1 2	5.0 6.0 6.0	165
 Urea	 Chlorocholine chloride	2 : 1	MCC (MW = 3.12 × 10^5 Da)	100 80[a] 100[b]	10 1 2	5.0 5.0 3.0	165
 Urea	 Betaine hydrochloride	4 : 1	MCC (MW = 3.12 × 10^5 Da)	100 80[a]	10 1	2.5 2.5	165
 Glycerol	 Choline chloride	2 : 1	MCC (MW = 3.12 × 10^5 Da)	100 80[a] 100[b]	10 1 2	3.0 2.5 3.5	165

Table 4 (*continued*)

HBD	HBA	Ratio	Source of cellulose?	T [°C]	t [h]	Sol. [wt%]	Ref.
 Thiourea	 Choline chloride	2 : 1	MCC (MW = 3.12 × 10^5 Da)	100 80[a] 100[b]	10 1 2	10.0 6.0 7.0	165

[a] Ultrasonic treatment. [b] Microwave treatment. [c] Not addressed.

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Table 5 Ionic liquids able to dissolve commercially available lignin

Ionic liquid	T [°C]	Sol. [w%]	Lignin source	Ref.
[C_4mim][PF_6]	120	0	Softwood Kraft	168
[Bmpyrr][PF_6]	120	0	Softwood Kraft	168
[C_4mim][BF_4]	100	12	Softwood Kraft	168
[C_1mim][$MeSO_4$]	90	50	Indulin AT	169
[C_4mim][CF_3SO_3]	90	50	Indulin AT	169
[Amim][Cl]	90	30	Indulin AT	169
[C_2mim][CH_3COO]	90	30	Indulin AT	169
[C_2C_2IM][$MeCO_2$]	90	30	Indulin AT	170
DMEAF	90	28	Indulin AT	170
DMEAA	90	19	Indulin AT	170
DMEAG	90	17	Indulin AT	170
DMEAS	90	10	Indulin AT	170
[C_4mim][Cl]	90	10	Indulin AT	170
[Bzmim][Cl]	90	10	Indulin AT	169
[C_4mim][Cl]	90	10	Indulin AT	169
[C_4mim][Br]	90	10	Softwood Kraft	169
[C_4mim][BF_4]	90	4	Indulin AT	169
[C_4mim][PF_6]	90	1	Indulin AT	169
[C_4mim][PF_6]	90	1	Softwood Kraft	170
[Bmmim][BF_4]	75	14	Softwood Kraft	168
[C_4mim][Cl]	75	13	Softwood Kraft	168
[C_6mim][CF_3SO_3]	70	22	Softwood Kraft	168
[C_4mim][$MeSO_4$]	50	26	Softwood Kraft	168
[C_2C_2IM][$MeSO_4$]	50	26	Softwood Kraft	168
[C_2C_2IM][$MeSO_4$]	25	6	Softwood Kraft	168
[C_4mim][$MeSO_4$]	25	6	Softwood Kraft	168
[Py][CH_3COO]	90	>50	Kraft lignin	167
[C_1mim][CH_3COO]	90	>50	Kraft lignin	167
[Pyrr][CH_3COO]	90	>50	Kraft lignin	167

valuable information, the solubility of such model compounds for lignin in an IL does not guarantee that the same IL will also excel at extraction of native lignin from a biomass substrate. For this reason, several studies were performed aiming for the reduction of lignin content in real lignocellulosic biomass using ILs. It should be mentioned that for the extraction of lignin, cellulose, and starch from biomass, the H-bonding between solute and solvent molecules must be strong enough to break down inter- and intramolecular H-bonds in the biomass. Only ILs with anions with a strong H-bond accepting ability, *e.g.* halides and acetate, were found to work well.

In a recent study the ILs [C_2CNBzim][Cl], [C_2CNAim][Cl] and [C_4mim][Cl] were found to be effective in extracting lignin with 53%, 47% and 38% lignin extraction, respectively.[180] Furthermore, a mixture of the IL [C_2mim][CH_3COO] and water was demonstrated to be effective for the pretreatment of lignocellulosic biomass, removing 75 wt% of the lignin.[181] The most effective mixture so far was found to be the combination of monoethanol (MEA), sulfurdioxide (SO_2), and 1,8-diazabicyclo-[5.4.0]-undec-7-ene (DBU). This combination was depicted as MEA-SO2-SIL,

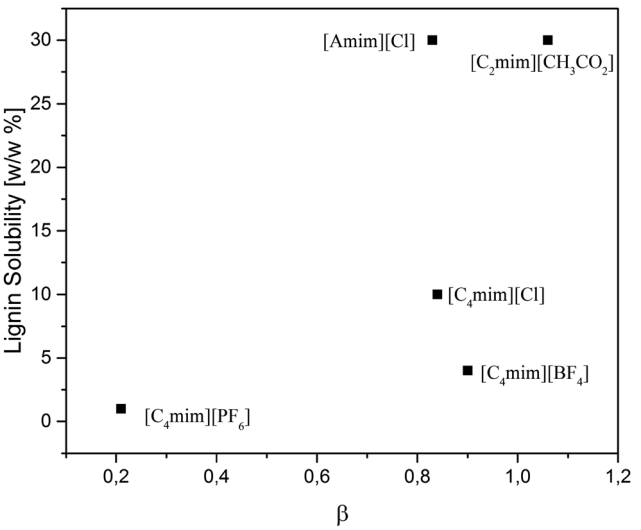


Fig. 8 Lignin solubility over the Kamlet–Taft β parameter.[148,178,179]

where SIL stands for a switchable ionic liquid. The results showed 80 wt% lignin extraction.[182]

The cation species has an effect on the extraction efficiency similar to that on lignin dissolution. ILs that have cations with an electron-rich aromatic π-system have been observed to produce stronger interactions with polarizable solute molecules, such as lignin.[183,184] The extraction yield of phenolic compounds proved to be a little higher for [BPy][Cl] than for [C_4mim][Cl] and the lowest yield was observed for [$(CH_3)_4$N][Cl]. This can be explained by the more aromatic character of $[BPy]^+$ in comparison with $[C_4mim]^+$.[185] A longer alkyl chain length of the cation enhances the extraction yield of lignin, as its hydrophobicity increases with increasing alkyl chain length. Apart from the hydrophobicity, the viscosity increases. Hence, the longer chain length benefits the extraction of some middle to less polar compounds, while the increase in viscosity limits their diffusion.[185–188]

A method to overcome the limits caused by viscosity is the usage of a co-solvent. For example, the addition of DMSO has proven to be beneficial for the extraction of lignin from wood, increasing the extraction efficiency by almost 50%.[189] The improvement in extraction yield in the presence of DMSO is only partially due to decreased viscosity, since it also has the potential to loosen the hydrogen bond network of the wood cellulosics, enhancing the penetration of the IL.

Another studied co-solvent is water, mainly since the presence of water is often inevitable during biomass treatments. Small amounts of water, <0.5 wt%, in the IL do not have a high impact

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

on the extraction efficiency.[189] Higher amounts decrease the viscosity significantly, but are also detrimental for extraction yields. In fact, water at high concentrations is an anti-solvent for lignin. Sun *et al.* completely dissolved wood in [C_2mim][CH_3COO], isolating up to 0.3 mass fraction of the initial wood lignin by selective precipitation in a mixture of acetone and water.[189]

5.5. Lignin dissolution and extraction from lignocellulosic biomass with deep eutectic solvents

In the search for an efficient and non-destructive process for lignocellulosic biomass fractionation, DESs have shown the first promising steps towards this goal.[65] In 2012 some DESs were evaluated for biopolymer dissolution, showing remarkably high lignin solubility compared to cellulose.[65] Especially DESs with lactic acid as HBD show high selective solubility, up to 12.03 w%, for lignin and no detectable solubility for cellulose (Table 6). Moreover, the DES consisting of malic acid and proline showed high lignin solubility. This suggests that DESs could very well be employed to design a selective lignin extraction from lignocellulosic biomass.

Kumar *et al.* recently reported extracting up to 60 ± 5% of the total lignin in rice straw biomass with a maximum of 10% solids loading.[190] The DES based delignification of this rice straw had no significant effect on the cellulose and hemicellulose fractions.[190]

Procentese *et al.* have reported an 88 wt% decrease of acid insoluble lignin of corncob biomass samples after treatment at 150 °C and 43 wt% at 80 °C with a choline chloride–imidazole DES.[191] The crystallinity index of the biomass sample was measured before and after treatment, and was found to be unchanged for treatment at 80 °C, which implies that the DES pretreatment resulted in a reduction of the cellulose crystallinity, since the amorphous lignin has been reduced.

Addition of small amounts of water during pretreatment significantly enhanced the total lignin extraction, and nearly 22 ± 3% more lignin was released from the residual biomass into the DES extract. However, additional results showed that water contents above 5% (w/v) affected the lignin extractability negatively.[190] An overview of the lignin extractability using DES is presented in Table 7.

5.6. Lignocellulosic biomass dissolution and regeneration with ionic liquids

The research into the dissolution of lignocellulosic biomass with ILs started in 2007 when two publications regarding this topic arose in the literature. The first publication showed partial dissolution of 5 wt% wood chips in the mixture of the AIL [C_4mim][Cl] and the co-solvent DMSO-d_6 at 100 °C.[192] The types of woods used in this investigation were pine, poplar, eucalyptus and oak, and dissolution times varied from 2 to 24 h. It was shown that dissolution was easier for softwoods than for hardwoods. To investigate the dissolved wood, it was regenerated with either a mixture of water and acetone (1 : 1), dichloromethane, or acetonitrile. The other publication in 2007 used [C_4mim][Cl], [Amim][Cl], 1-methyl-3-benzylimidazolium chloride ([benzylmim][Cl]), 1-methyl-3-*m*-methoxylbenzylimidazolium chloride ([methoxybenzylmim][Cl]), 1-methyl-3-methylbenzylimidazolium chloride ([methylbenzylmim][Cl]) and 1-methyl-3-benzyl-imidazolium 1-methyl-3-benzyl-imidazolium dicyanamide ([benzylmim][Dca]) for testing the dissolution of unbleached Norway spruce thermomechanical pulp (TMP), Southern pine TMP, Norway spruce dust and ball-milled Southern pine powder and wood chips.[174] The results demonstrated that especially [C_4mim][Cl] and [Amim][Cl] had high ability for dissolving the wood-based lignocellulosic biomass. The best results were obtained for the dissolution of Norway spruce sawdust at heating to 100 °C for 8 h, with 8 wt% dissolution for both [C_4mim][Cl] and [Amim][Cl]. Moreover, 8 wt% of ball-milled Southern pine powder could be dissolved with the AIL [Amim][Cl]. The dissolved wood was regenerated with distilled water.

Table 6 Solubility of lignin and cellulose in various DESs[65]

HBD	HBA	Ratio	T [°C]	Lignin [wt%]	Cellulose [wt%]
Lactic acid	Proline	2 : 1	60	7.56	0.00
Lactic acid	Betaine	2 : 1	60	12.03	0.00
Lactic acid	Choline chloride	1.3 : 1	60	4.55	0.00
Lactic acid	Choline chloride	2 : 1	60	5.38	0.00
Lactic acid	Choline chloride	5 : 1	60	7.77	0.00
Lactic acid	Choline chloride	10 : 1	60	11.82	0.00
Lactic acid	Histidine	9 : 1	60	11.88	0.00
Lactic acid	Glycine	9 : 1	60	8.77	0.00
Lactic acid	Alanine	9 : 1	60	8.47	0.00
Malic acid	Alanine	1 : 1	100	1.75	0.11
Malic acid	Betaine	1 : 1	100	0.00	0.00
Malic acid	Choline chloride	1 : 1	100	3.40	0.00
Malic acid	Glycine	1 : 1	100	1.46	0.14
Malic acid	Proline	1 : 1	100	0.00	0.00
Malic acid	Proline	1 : 2	100	6.09	0.24
Malic acid	Proline	1 : 3	100	14.90	0.78
Malic acid	Histidine	2 : 1	85	0.00	0.00
Malic acid	Nicotinic acid	9 : 1	85	0.00	0.00
Oxalic acid[a]	Betaine	1 : 1	60	0.66	0.00
Oxalic acid[a]	Proline	1 : 1	60	1.25	0.00
Oxalic acid[a]	Choline chloride	1 : 1	60	3.62	0.00
Oxalic acid[a]	Glycine	3 : 1	85	0.28	0.00
Oxalic acid[a]	Nicotinic acid	9 : 1	60	0.00	0.00
Oxalic acid[a]	Histidine	9 : 1	60	0.00	0.25
Oxalic acid[b]	Choline chloride	1 : 1	60	0.00	0.00
Oxalic acid[b]	Proline	1 : 1	60	0.00	0.15

[a] The dihydrate form of oxalic acid was used. [b] The anhydrous form of oxalic acid was used.

Table 7 Extractability of lignin of various DESs

HBD	HBA	Ratio	T [°C]	Extracted [wt%]	Ref.
Glycerol	Choline chloride	2 : 1	80	4.4	191
Glycerol	Choline chloride	2 : 1	115	8.8	191
Glycerol	Choline chloride	2 : 1	150	24.8	191
Urea	Choline chloride	2 : 1	80	8.0	191
Urea	Choline chloride	2 : 1	115	24.8	191
Imidazole	Choline chloride	7 : 3	80	43.1	191
Imidazole	Choline chloride	7 : 3	115	70.8	191
Imidazole	Choline chloride	7 : 3	150	88.3	191
Lactic acid	Betaine	2 : 1	60	52.0	190
Lactic acid	Betaine	5 : 1	60	56.0	190
Lactic acid	Choline chloride	2 : 1	60	51.0	190
Lactic acid	Choline chloride	5 : 1	60	60.0	190
Lactic acid	Choline chloride	9 : 1	60	59.0	190

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

In 2009 investigations showed several other AILs able to dissolve wood. The AILs [Amim][Cl], [C_2mim][Cl], [C_4mim][Cl], [ECOENG][1111P] and [C_2mim][CH_3COO] were tested for the dissolution of 5 wt% wood chips of the species spruce, silver fir, common beech and chestnut at 90 °C, while shaking for 12 h.[175] The results indicated that [C_2mim][Cl], [C_4mim][Cl] and [ECOENG][1111P] partially dissolved all the wood species, while [C_2mim][CH_3COO] could completely dissolve three of the four wood chips except for the silver fir, which was only partially dissolved. All four types of wood chips could be dissolved with the AIL [Amim][Cl]. It was hypothesized that the π-electrons of the side chain of the cation play an important role, since the cation has the capability to undergo π–π interactions with the aromatic compounds of lignin contributing to the dissolution ability of this AIL.[175] Furthermore, the low viscosity could play a role. Another investigation in 2009 aimed at full dissolution of lignocellulosic biomass with [C_2mim][CH_3COO] in comparison to [C_4mim][Cl].[193] [C_2mim][CH_3COO] was chosen for its higher hydrogen bond basicity, and was hypothesized to be a more effective solvent for dissolving the lignocellulosic biomass. A softwood, southern yellow pine, and a hardwood, red oak, were used for the investigation. The results demonstrated that red oak can be fully dissolved within 25 h and southern yellow pine within 46 h, both at 110 °C. It was observed that smaller particle sizes led to easier dissolution, which can be explained by the increased surface area and the breaking down of the structure due to the pre-treatment.

Continued investigations showed that viscosity, water content and temperature played an important role in the swelling and the partial dissolution of pine sapwood.[194] Kamlet–Taft parameters were investigated to correlate the dissolution behaviour. The results indicated that especially the Kamlet–Taft β parameter needs to be higher than 0.8.

In 2011 research continued with the dissolution of different wood sorts with the aforementioned AILs. [Amim][Cl] in combination with 16 wt% of DMSO was used for the dissolution of approximately 5 wt% of pine, poplar, Chinese parasol or catalpa wood samples at different temperatures.[195] Since a colour change and an increase of viscosity were observed, it was assumed that wood was fully dissolved. As found before, softwood, in this case pine, can be easily dissolved than the hardwood. The authors further indicated that a reduction of dissolution time was possible by the use of microwave irradiation instead of heating in an oil bath. In another publication the disintegration and dissolution kinetics of wood chips, beech and spruce wood, were investigated for the AIL [C_2mim][CH_3COO].[196] Wood particles varying from 0.1 to 0.5 mm were used. The results showed that approximately 40% of spruce and 75% of the beech could be dissolved at 115 °C. No full dissolution was observed after 72 h. It was observed that the kinetics can be considered as a first order reaction. An interesting research study showed the dissolution of bagasse within 5–15 min at temperatures up to 195 °C and approximately 90% dissolution of southern yellow pine at 175 °C after 30 min.[197] These dissolution temperatures are both above the glass transition temperature of lignin. Despite the major decrease of the dissolution times, about 15% of AIL disintegrated, which contributes to a lower overall efficiency when recycling is taken into account. Later in 2011, a combination of acetic acid with the ILs [C_2mim][CH_3COO], [C_2mim][Cl] and [C_4mim][Cl] was used for the dissolution of wheat straw and pine wood.[6] It was shown that not only dissolution but also hydrolysis occurs.

It was also tried to dissolve *Pinus radiata* (0.35 g) with the AIL [C_2mim][CH_3COO] (7.0 g) at temperatures of 120 °C and 155 °C.[198] It was observed that after 3 h the wood particles swelled and partially dissolved, but no complete dissolution was achieved. In 2012, 4 wt% of *Pinus radiata* and *Eucalyptus globulus* was dissolved in the IL [AMIM][Cl].[199] The extractives from these woods were removed with acetone and water, after which dissolution was conducted in a microwave oven first by heating at 110 °C for 10 min and then by heating at 120 °C for 20 min. After dissolution, DMSO was added to decrease the viscosity of the system, after which the mixtures were filtered to remove the parts that were not dissolved (34%). The solutions were added to 200 mL of methanol to precipitate the cellulose *via* stirring at 300 rpm for 10 min and at a temperature of 40 °C. FTIR analysis showed that the precipitated cellulose was comparable to MCC, while XRD and ^{13}C NMR proved that crystallinity of the regenerated cellulose decreased due to the dissolution and regeneration processes. In 2013 this research was continued with the same two sorts of wood, but in this case both raw wood and extractive-free wood were used.[200] The solutions of the mixtures of wood and IL were first heated at 110 °C for 10 min, after which the temperature was increased to 120, 140 or 170 °C for approximately 20, 40 or 60 min. Methanol was used as an anti-solvent to regenerate cellulose and lignin. It was found that by varying the regeneration temperature and time, either pure cellulose, both cellulose and lignin, or pure lignin can be regenerated. Later in that year, three methods were described for regeneration.[201] A brief summary is depicted, but for a broad description the reader is redirected to the original publication.[201] The 1st method was based on the dissolution of 250 mg of wheat straw particles (<0.5 mm) in 5 g of the IL [C_2mim][CH_3COO] for 6 h under continuous stirring and heating at 120 °C.[201] After the dissolution, 0.1 M NaOH was used for the precipitation of the carbohydrate material. This material was collected by filtration and washed with ultrapure water, after which the lignin was precipitated from the mixture with the addition of 1 M HCl and heating the mixture up to 70 °C during 30 min. The lignin was washed with 10 mL of ultrapure water and both the lignin and the carbohydrate materials were dried for 24 h at 60 °C. The 2nd method describes the dissolution of 100 mg of the before mentioned wheat straw in 5 grams of [C_2mim][CH_3COO] for 4 h at a temperature of 110 °C.[201] In this method regeneration is performed upon addition of a mixture consisting of acetone/water (9 : 1, v/v), after which the solution was centrifuged at 4000 rpm for 15 min at 22 °C. The the carbohydrate-rich phase was filtered and then washed again in a 1 : 1 (v : v) mixture of acetone/water after which it was centrifuged at 9000 rpm for 30 min at 4 °C following which the carbohydrates were dried for 18 h at 60 °C. From the lignin containing phase acetone was removed *via* reduced pressure, after which the pH was lowered to 2.0 upon addition of 1 M HCl. This step was conducted to

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

precipitate lignin-rich material, which was subsequently washed with 0.01 M HCl. The third method was an optimized regeneration by combining the two previous mentioned methods.[201]

Another publication in 2013 presented ILs based on amino acids.[202] It was observed that the IL *N*-methyl-*N*-(2-methoxyethyl)-pyrolidin-1-ium 2,6-diaminohexanoate ([P1ME][Lys]) was able to dissolve lignin at 60 °C and not cellulose, while both could be dissolved at 80 °C. With this IL, the authors extracted lignin from Japanese cedar (*Cryptomeria japonica*). They added 1.0 g of the lignocellulosic biomass to the IL and stirred it for 12 h at 60 °C. The lignin precipitated upon addition of methanol or ethanol. In general, this IL could even be used to fully dissolve the lignocellulosic biomass at higher temperatures, after which cellulose could be precipitated upon decreasing the temperature of the mixture. Furthermore, cellulose originating from ammonia fiber expansion (AFEX) pretreated *Zoysia japonica* could be dissolved.[203] It was shown that AFEX pretreatment reduces the amount of lignin and hemicellulose in the biomass and thus breaks the structure of the lignocellulose. For the dissolution, 2 g of the pretreated *Zoysia japonica* was dissolved in 50 g of the IL [Amim][Cl]. Moreover, the results indicated that approximately 97% of the cellulose could be regenerated. Wood was also pretreated with autohydrolysis to increase the fractionation of the birchwood.[204] The results showed that with the pretreatment 15 wt% of the birchwood could be dissolved, while this was unrenderable without the pretreatment. Upon precipitation, it was shown that the cellulose-to-lignin ratio changed from 2.7 to 5.7 (comparison of non-treated with pre-treated). A more fundamental research study gave physical insight into the dissolution of switchgrass in the IL [C_2mim][CH_3COO] with small-angle neutron scattering.[205] The results showed that although dissolution of the cellulose fibrils occurs, the supramolecular network of the biopolymer stays intact. However, this is not present in solutions of the individual biopolymers, and thus, no self-assembly occurs. Finally, a recent paper showed the complete dissolution of switchgrass in [C_4mim][Cl] *via* ultrasonification.[206] An overview of all ionic liquids that have been used to dissolve lignocellulosic biomass can be found in Table 8.

6. Other extractables

Apart from the three most obvious lignocellulosic biomass constituents – lignin, cellulose and hemicellulose – alternative biomass components can be valorised. These components are typically located outside the plant's stem, for instance in fruits, seeds or leaves. Attempts have been made to process, isolate and/or valorise these components with both ILs and DESs. The treatment of plant materials with ILs has been extensively reviewed in the literature;[207,208] therefore the focus of the following sections will be on DESs. Consequently, the performance of DESs will be compared to that of ILs for each particular application where both solvent categories have been studied.

6.1. Starch processing

Together with (hemi)celluloses, starch is an omnipresent polysaccharide found in vegetal sources. Several studies present notable starch solubility in DESs (see Table 9), while starch is poorly soluble in water and most other polar solvents.[209] Since DESs exhibit strong affinities with starch, they were explored for the plasticisation of thermoplastic starch. A DES and starch were introduced to an extruder resulting in plastic pallets that can be moulded into films. Plasticisers (conventionally glycerol, urea, ethanolamine or formamide) are added to starch films to decrease some of its weaknesses, such as recrystallization and moisture absorption, both leading to reduced mechanical properties.[210,211] Starch plasticised with glycerol : ChCl 2 : 1 resulted in a colourless translucent film, while the incorporation of urea/ChCl based DESs resulted in a transparent slightly yellow film with better material properties.[210,211]

In addition to DESs, ILs can act as plasticisers at low concentrations (<25 wt%). Similarly, the inclusion of [Amim][Cl] or [C_4mim][Cl] lowers the hygroscopicity of the starch film and improves its material properties compared to that achieved with the conventional glycerol.[213] An advantage of [C_4mim][Cl] over the tested DESs and glycerol is that it allows the starch films to be compatible with zein.[214] Zein is a hydrophobic protein that is abundant in corn. This allows the production of films from corn starch with higher zein impurities, reducing the level of corn flour refining necessary. The two tested ChCl containing DESs showed less compatibility with zein than [C_4mim][Cl].[214]

Another advantage of using ILs as starch plasticisers over DESs is that the resulting film has a relatively high electrical conductivity.[213] Polymer electrolytes made with IL plasticized starch have conductivities in the range of 10^{-5} to $10^{-1.5}$ S cm^{-1}, while DES based starch films show conductivities in the order of 10^{-7} S cm^{-1}.[207,210,213] The advantages of the DES (cheap, easy preparation) were combined with the advantages of lithium bis(trifluoromethanesulfonyl)imide (LITFSI) plasticised starch. The conductivities of these DES-LITFSI-starch films were in the order of 10^{-5} to 10^{3}.[215] However, not many details were given on other material properties. A more elaborate discussion on ILs and starch plasticisation, dissolution and modification can be found in an earlier review.[207]

6.2. Plant extractives

Two main plant material valorisation routes can be distinguished: (1) utilising the pervasive biopolymers, isolate and/or modify them into moderate added-value materials or chemicals and (2) isolating specific, low amounts of valuable components present in the biomass. The last route creates an extensive added-value to the biomass and circumvents the often costly synthesis necessary to obtain the components artificially. However, it requires an extractant with high selectivity towards the solute. Designer solvents like DESs and ILs can offer this feature. A detailed overview of viable extraction pathways for value-added compounds originating from biomass utilising ILs is given in a recent review.[208]

In 2013, the extraction, stabilisation and recovery of yellow and red pigments from safflower with DESs were studied.[216,217] It was found that the main hurdle to successful extraction was the DESs' inherent high viscosity. Increasing the extraction temperature from ambient temperature to 40 °C could decrease the viscosity of a DES significantly.[68,82] However, the biggest

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Table 8 Ionic liquids that can dissolve lignocellulosic biomass

IL	Biomass type	Recovery solvent(s)	T [°C]	t [min]	Particle size [mm]	Ref.
[C_4mim][Cl][a]	Pine	1 : 1 acetone–water	100	120	5	192
	Poplar	Dichloromethane		360		
	Eucalyptus	Acetonitrile		720		
	Oak			1440		
[C_4mim][Cl]	Wood chips	Distilled water	130	900	N.A.	174
[C_4mim][Cl]	Norway spruce sawdust	Distilled water	110	480	0.1–2	174
[C_4mim][Cl]	Southern pine TMP	Distilled water	130	480	N.A.	174
[Amim][Cl]	Ball-milled Southern pine powder	Distilled water	110	480	N.A.	174
[Amim][Cl]	Norway spruce sawdust	Distilled water	110	480	0.1–2	174
[Amim][Cl]	Norway spruce dust	Distilled water	80	1440	0.1–2	174
[Amim][Cl]	Norway spruce TMP	Distilled water	130	480	N.A.	174
[Amim][Cl]	Southern pine TMP	Distilled water	110	480	N.A.	174
[Amim][Cl]	Southern pine TMP	Distilled water	130	480	N.A.	174
[bzmim][Cl]	Southern pine TMP	Distilled water	130	480	N.A.	174
[bzmim][Cl]	Norway spruce TMP	Distilled water	130	480	N.A.	174
[bz-ome-mim][Cl]	Southern pine TMP	Distilled water	130	480	N.A.	174
[benzylmim][Dca]	Southern pine TMP	Distilled water	130	480	N.A.	174
[Amim][Cl]	Spruce	N.A.	90	1080	1–2	175
	Silver fir					
	Common beech					
	Chestnut					
[C_2mim][Cl]	Spruce	N.A.	90	1080	1–2	175
	Silver fir					
	Common beech					
	Chestnut					
[C_4mim][Cl]	Spruce	N.A.	90	1080	1–2	175
	Silver fir					
	Common beech					
	Chestnut					
[ECOENG][1111P]	Spruce	N.A.	90	1080	1–2	175
	Silver fir					
	Common beech					
	Chestnut					
[C_2mim][CH_3COO]	Spruce	N.A.	105	1080	1–2	175
	Silver fir					
	Common beech					
	Chestnut					
[C_2mim][CH_3COO]	Southern yellow pine	Acetone/water (1 : 1 v/v)	110	2760	0.125	197
	Red oak			1500	0.125–0.250	
					0.250–0.500	
					0.500–1.000	
[C_4mim][OTf]	Pine	N.A.	60–120	$\leq$60 h	10 × 10 × 5	194
[C_4mim][$N(CN)_2$]						
[C_4mim][$MeSO_4$]						
[C_4mim][Cl]						
[C_4mim][Me_2PO_4]						
[C_4mim][$MeCO_2$]						
[Amim][Cl]	Pine	DSMO/water	90	min–h	0.45–0.65	195
	Poplar		120			
	Chinese parasol					
	Catalpa					
[C_2mim][CH_3COO]	Spruce	N.A.	115	$\leq$72 h	0.1–0.5	196
	Beech					
[C_2mim][CH_3COO]	Bagasse	Acetone : water (1 : 1 v/v)	110	900–960	$<$0.125	197
[C_2mim][CH_3COO]	Bagasse	Acetone : water (1 : 1 v/v)	175–195	5–15	$<$0.125	197
[C_2mim][CH_3COO]	Pine	Acetone : water (1 : 1 v/v)	175	30	$<$0.25	197
[C_2mim][CH_3COO]	Wheat straw	Water	100	1440	$<$0.1	6
	Pine					
[C_2mim][Cl]	Wheat straw	Water	100	1440	$<$0.1	6
	Pine					
[C_4mim][Cl]	Wheat straw	Water	100	1440	$<$0.1	6
	Pine					
[C_2mim][CH_3COO][b]	Wheat straw	Ethanol (cellulose)	100	300	$<$0.1	6
	Pine	Water (lignin)	125			
[C_2mim][Cl][b]	Wheat straw	Ethanol (cellulose)	100	300	$<$0.1	6
	Pine	Water (lignin)	125			
[C_4mim][Cl][b]	Wheat straw	Ethanol (cellulose)	100	300	$<$0.1	6
	Pine	Water (lignin)	125			
[C_2mim][OAc]	*Pinus radiata*	Water	120	180	0.420–0.841	198
			155			

This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

Table 8 (*continued*)

IL	Biomass type	Recovery solvent(s)	T [°C]	t [min]	Particle size [mm]	Ref.
[Amim][Cl]	*Pinus radiata* and *Eucalyptus globulus* woods	Methanol	110–120	10 20	N.A.	199
[Amim][Cl]	*Pinus radiata* and *Eucalyptus globulus*	Methanol	110–120 140 170	10 20 40 60	N.A.	200
[C_2mim][CH_3COO]	Wheat straw	0.1 M NaOH 1.0 M HCl NaOH pelleter acetonitrile	120	360	<0.5 mm	201
[C_2mim][CH_3COO]	Wheat straw	Acetone/water (9 : 1, v/v) Acetone/water (1 : 1, v/v) Ultrapure water 1.0 M HCl 3% (w/w) NaOH 4 M HCl 96% (v/v) ethanol	110	240	<0.5 mm	201
[C_2mim][CH_3COO]	Wheat straw	0.1 M NaOH distilled water HCl 4 M HCl 1 M 96% (v/v) ethanol HCl 0.02 M 3% (w/w) NaOH	120	360	<0.5 mm	201
[C_2mim][Cl]	*Zoysia japonica*	Deionized water	80	30	<0.7 mm	203
[C_2mim][CH_3COO]	Birchwood	Acetone/water (1 : 1 v/v)	100	N.A.	<0.125 mm	203 and 204
[C4mim][Cl]	Switchgrass	Acetone Water	130 100–150[b]	720–1440 2–4	N.A.	206

[a] Not addressed. [b] Ultrasonification.

Table 9 Starch solubility in several ILs and DESs

Solvent	Sol [wt%]	T [°C]	t [h]	Ref.
[Amim][Cl]	15	80	0.66	212
	20	100	15	
	50	100	n.r.	
[C_4mim][Cl]	15.0	80	n.r.	209
$ZnCl_2$: ChCl 1.9 : 1	4.6	98	n.r.	209
Urea : $CaCl_2$ 4 : 1[a]	16.7	80	n.r.	209
Urea : ChCl 1 : 1[a]	9.1	100	n.r.	209
Citric acid : ChCl 1.4 : 1[a]	8.3	100	n.r.	209
Oxalic acid : ChCl 1 : 1.6[a]	9.1	100	n.r.	209
Oxalic acid : ChCl 1 : 1	0.15	60	<24	65
Glucose : ChCl 1 : 1	17.2[b]	n.r.	n.r.	56
Lactic acid : ChCl 10 : 1	0.13	60	<24	65
Lactic acid : histidine 9 : 1	0.13	60	<24	65
Lactic acid : alanine 9 : 1	0.26	60	<24	65
Malic acid : alanine 1 : 1	0.29	100	<24	65
Malic acid : betaine 1 : 1	0.81	100	<24	65
Malic acid : ChCl 1 : 1	7.10	100	<24	65
Malic acid : glycine 1 : 1	7.65	100	<24	65
Malic acid : proline 1 : 2	0.32	100	<24	65
Malic acid : proline 1 : 3	5.90	100	<24	65
Oxalic acid : nicotinic acid 9 : 1	2.83	60	<24	65
Oxalic acid : proline 1 : 1	0.15	60	<24	65

[a] The ratios given are mass ratios instead of molar ratios. [b] The unit used is mg ml^{-1} rather than wt%.

reduction of viscosity was obtained by the addition of 10–25 wt% water. Extraction yields of the diluted DESs were analysed for polar and less polar pigments at multiple water contents. The optimum DES–water mixtures displayed comparable or better extraction results than the reference solvents water, ethanol and the 60/40 (v/v) water/ethanol mixture.[216] This emphasises once more the tunability of DESs and their compatibility with water. The stability and solubility of the pigment could also be controlled by the DESs' water content: decreasing the water content yielded improved stability and the solubility attained an optimum at 10% (v/v) water.[68,217]

The same strategy was applied for the extraction of compounds from traditional Chinese medicines (TCM), which usually originate from plant roots, leaves or blossoms. Recently, the extraction of bioactive compounds from pigeon pea roots (*Cajanus Cajan*)[218] and sophora flowers (*Flos Sophorae*),[219] flavonoids from skullcap root (*Radix Scutellariae*)[220] and marsh horsetail (*Equisetum Palustre*)[221] and phenolics from *Pyrola incarnata Fisch*[222] and dried sprouts of the redstem wormwood (*Herba Artemisiae Scopariae*)[223] was studied. The aim of the extraction experiments is to either isolate or quantify the active components that are present in the often complex matrices of TCMs. The extraction efficiencies of the DESs were optimised through statistical analysis using microwave (MAE),[218,220,222] ultrasound (UAE)[218,219] and negative pressure cavitation (NPCAE)[221] assisted extraction. The optimisation process typically focuses on a series of single factor experiments. Herein, the DESs' components, ratio and water content are optimised followed by the extraction conditions (temperature, time, sample loading). In all cases, the extraction process was more effective compared to using the common extractants, often (a mixture of water and) an alcohol.

In the aforementioned publications the performance of DESs was not directly compared to that of ILs, probably since the biomass source was not the same or because the optimisation process focussed on different techniques. Pigeon pea roots have, for instance, also been treated with ILs, but only using NPCAE.[224] Lab-scale DES-MAE could obtain yields of 0.449, 0.617 and 0.221 mg $g_{dryweight}^{-1}$ for genistin, genistein and apigenin,[218] respectively, while the yields for those flavonoids were 0.482, 0.496 and 0.291 mg $g_{dryweight}^{-1}$, respectively, for IL-NPCAE.[224] The yields are in the same order of magnitude, but since no

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

IL-MAE or DES-NPCAE studies were performed, it is questionable whether it is a fair comparison. For redstem wormwood no comparable direct extraction was performed with ILs, but a supported IL showed interaction with the phenolic acids that could be extracted with DESs.[223,225] This allows the alternative analysis of four bioactive components in a redstem wormwood extract.

Another analytical chemistry related application of designer solvents is the headspace single droplet microextraction (HSDME) using DESs and ILs. In this technique, a single micro droplet of a low-volatile solvent with a known volume is brought into contact with the headspace of a liquid or solid containing volatile compounds. The volatiles are extracted from the headspace into the micro droplet and the droplet is subsequently injected in a gas chromatograph for qualification or quantification of the volatiles.[226] Both DESs and ILs are described to have low volatility and could act as selective extractants for headspace samples.[227,228] The first application of DESs in a HSDME setup is the analysis of terpenoids in Japanese cypress (*Chamaecyparis obtuse*) leaves with ethylene glycol : ChCl mixtures.[227] Hence, in combination with HSDME, DESs can be used for the analysis of specific compounds in biomass samples.

DESs were also applied to the extraction of saponins from sisal waste and juá bark.[229] Saponins are plant metabolites that can be applied as bioinsecticides, surfactants and act as a basis for steroidal drugs.[229,230] ChCl based DESs and cholininium based ILs were mixed with water or a water/ethanol mixture and consecutively tested for saponin extraction.[229] The ILs generally performed slightly better than or similar to their corresponding DESs.[229] Nevertheless, the cholinium based ILs were omitted during extraction optimization since their estimated price is typically an order of magnitude higher than that of their corresponding DESs. After several optimisation steps, the most effective DES based extractants appeared to be the mixtures consisting of 58% propionic acid : ChCl 2 : 1 + 42% ethanol and 81% acetic acid : ChCl 2 : 1 + 19% water for juá bark and sisal waste, respectively. The DESs showed high selectivity and they could improve the extraction efficiency with a factor of 1.7 and 2.5 compared to ethanol and water, respectively. It has to be noted, however, that the pure acid constituents used in the DESs are liquid at room temperature and their eutectic behaviour was not reviewed.

By the use of ILs, other sources for saponins were explored. Dried leaves and aerial parts of mate (*Ilex paraguariensis*) and tea (*Camellia sinensis*) were treated with aqueous imidazolium and cholinium based ILs and with aqueous solutions of ChCl.[230] In another research study ginseng roots were subjected to aqueous imidazolium ionic liquid based ultrasonic assisted extraction experiments.[231] In both studies the influence of the imidazolium alkyl chain length on the extraction efficiency and selectivity was investigated. The alkyl chain length of the imidazolium cations affected the extraction performance, but the effect was different depending on the saponin source. [C_3mim][Br] clearly showed the highest extraction efficiency for ginseng roots, while for shorter and longer alkyl chains the efficiencies were significantly lower, although similar among them.[231] For tea, a different trend was observed: [C_2mim][Cl] and [C_4mim][Cl] had comparable extraction efficiencies, but a further increase in the alkyl chain length resulted in an obvious decrease in extraction efficiency. The extraction results of mate did not show a clear tendency with increasing cation chain length.[230] The influence of the extractants' anions on the extraction efficiencies of mate, tea, ginseng, juá bark and sisal waste was studied.[229–231] The anions of imidazolium based ILs did not appear to be vital for the extraction process, while for cholinium based ILs the efficiencies were strongly influenced by changing the anion. Again, different behaviours were observed for the distinct biomass sources.

It has to be noted that the saponins in tea, ginseng, juá bark, sisal waste and mate consist of different glycoside and terpene groups. This obviously influences the solute–solvent interactions.[229–231] Additionally, the solute accessibility changes with the biomass type and process conditions. This is expressed in the different temperature profiles for extraction efficiencies of saponins with ChCl from mate and tea. For mate the efficiencies increased significantly with temperature, while for tea they remained constant.[230] The size and polarity of the DESs' components and ILs' ions play different roles depending on the biomass variety. This highlights that a tailor-made extractant is often needed for the selective extraction of value-added components from biomass sources. Both DESs and ILs offer the versatility to exploit efficient, cheap and renewable extraction routes from biomass sources.

7. Challenges and outlook

The innovative solvents as discussed in this review show great promise for the processing of lignocellulosic biomass. With the current information available in the literature, AILs have the potential of overtaking the conventional methods used in the biorefinery industry, which is shown by their possibilities of large removal of lignin from the lignocellulosic biomass, up to 50 wt%, in combination with the large cellulose dissolution, up to 25 wt%. Next to their high extraction and dissolution capacity, various studies show their high thermal stability ensuring the usage of the AILs at the high temperatures needed for the extraction of lignin and the solvability of cellulose. Nevertheless, AILs also have some disadvantages, which should be resolved or further investigated before scale-up or implementation in industry is possible. Probably the biggest disadvantage of AILs is their production price. The combination of the low reaction yields and the use of conventional solvents leads to the need of extended purification steps increasing their price per ton. Thus, better procedures should be devised to decrease the production price. Next to their high production price, AILs often have moderate to high viscosities hampering their full potential due to a lower ability of dissolution caused by a large decrease of mass transfer. This problem could be overcome by the addition of conventional solvents such as DMSO. However, this introduces other problems related to toxicity and

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

the separation of these co-solvents from the AIL for their use in the regeneration.

Promising alternatives for the AILs are the PILs, which are able to dissolve cellulose (18 wt%) in combination with their ability to extract lignin (80 wt%) from lignocellulosic biomass. The major advantage of PILs over AILs is their easy preparation. While for the AILs a complete synthesis with extended purification is needed, only mixing of the acid and base is needed for the preparation of a PIL. A disadvantage of the preparation is the fact that it is exothermic, which introduces the need for cooling. At a lab scale this is a minor problem, but heat development can cause major problems during the upscaling process. Moreover, the thermal stability and the recycling of PILs should be further investigated. The ability of PILs to evaporate is often noted as a big advantage, and could be a method for recycling, but the boiling points are so high that enhanced temperatures would be needed. In our opinion, this will make applications such as distillation too energy-costly. Furthermore, only few publications have discussed the handling of cellulose and lignocellulosic biomass with PILs. Thus, in general it can be concluded that further investigation is needed before a concrete conclusion can be made, though until now they have shown great potential for a future in the biorefinery industry.

The last class of solvents that were discussed for their future use of treating lignocellulosic biomass and its constituents are the DESs. As mentioned before, DESs were only published for the first time in 2003. Thus, until now only a fraction of the work in the field of DESs has been done. Despite that, DESs already show great promise for a future use in the biorefinery industry. As published by our group in 2012, DESs can be used for dissolving up to 14.9 wt% lignin. More recent work has shown the potential of DESs when lignin was extracted from lignocellulosic biomass. It should be mentioned that the amounts of cellulose that can be dissolved are not high yet and in some cases the results are not reproducible. Therefore, at the moment the performance of DESs for biomass processing is still inferior to that of ILs. However, we think this is just a matter of time and finding the right functional groups and constituents. A major advantage of DESs is their easy preparation, which is by mixing their constituents at moderate temperatures. No cooling and no purification steps are needed, which makes them cheaper than ILs. Furthermore, the constituents of DESs are in most cases described as readily available and 'green'. We agree with this, although synergy/additivity effects should be investigated before calling a DES 'green'. A disadvantage already known in the literature is the viscosity of DESs, which is often rather high. Currently, there is also still a lack of fundamental knowledge about the formation of DESs and their interaction with solutes. Furthermore, more research should also be conducted into the thermal stability of the DESs to obtain a deeper insight into this topic.

For both the PILs and DESs the question whether only dissolution or also acid hydrolysis occurs remains. Up until now, only minor knowledge is present in the literature regarding the acidity of the PILs and the DESs. Thus, this should be addressed first before more discussion is possible.

In general, all three innovative solvents have potential for a future use in the biorefinery for the valorisation of lignocellulosic biomass and the alternative biomass constituents. However, scale up experiments should be conducted to investigate their full potential. Questions that still arise from the literature are: (i) How is the recyclability of these innovative solvents? (ii) What is the influence of water on their performance? (iii) Are the particular solvents proposed for use in the biorefinery industry as 'green' as thought? (iv) Can they be produced on the large scale needed for the bio-refinery (ton/year scale)?

Acknowledgements

This project has received funding from the Bio-Based Industries Joint Undertaking under the European Union's Horizon 2020 research and innovation programme under grant agreement No. 668970. The authors would like to thank the ISPT cluster ''Deep Eutectic Solvents in the pulp and paper industry'' consortium for their financial and in kind contribution. This cluster consists of the following organisations: Altri – Celbi, Buckman, Crown Van Gelder, CTP, DS Smith Paper, Eska Graphic Board, Holmen, ISPT, Mayr-Melnhof, Metsä Fibre, Mondi, Omya, Parenco BV, Portucel, Sappi, Smurfit Kappa, Stora Enso, Eindhoven University of Technology, UPM, Van Houtum, Voith Paper, VTT Technical Research Centre of Finland Ltd and Zellstoff Pöls. Furthermore, this research is supported by Paques B. V. and the Dutch Technology Foundation STW, which is part of the Netherlands Organisation for Scientific Research (NWO) and which is partly funded by the Ministry of Economic Affairs.

Notes and references

1 P. Kumar, D. M. Barrett, M. J. Delwiche and P. Stroeve, *Ind. Eng. Chem. Res.*, 2009, **48**, 3713–3729.
2 C. O. Tuck, E. Pérez, I. T. Horváth, R. A. Sheldon and M. Poliakoff, *Science*, 2012, **337**, 695–699.
3 A. Demirbas, *Energy Convers. Manage.*, 2001, **42**, 1357–1378.
4 S. Zhu, Y. Wu, Q. Chen, Z. Yu, C. Wang, S. Jin, Y. Ding and G. Wu, *Green Chem.*, 2006, **8**, 325–327.
5 G. W. Huber, S. Iborra and A. Corma, *Chem. Rev.*, 2006, **106**, 4044–4098.
6 J. van Spronsen, M. A. T. Cardoso, G.-J. Witkamp, W. de Jong and M. C. Kroon, *Chem. Eng. Process.*, 2011, **50**, 196–199.
7 H. Wang, G. Gurau and R. D. Rogers, *Chem. Soc. Rev.*, 2012, **41**, 1519–1537.
8 A. Brandt, J. Gräsvik, J. P. Hallett and T. Welton, *Green Chem.*, 2013, **15**, 550–583.
9 N. Sun, H. Rodríguez, M. Rahman and R. D. Rogers, *Chem. Commun.*, 2011, **47**, 1405–1421.
10 A. C. O'Sullivan, *Cellulose*, 1997, **4**, 173–207.
11 Y.-B. Huang and Y. Fu, *Green Chem.*, 2013, **15**, 1095–1111.
12 S. Morales-delaRosa, J. M. Campos-Martin and J. L. Fierro, *Chem. Eng. J.*, 2012, **181**, 538–541.
13 R. R. Singhania, A. K. Patel, R. K. Sukumaran, C. Larroche and A. Pandey, *Bioresour. Technol.*, 2013, **127**, 500–507.
14 D. Chiaramonti, M. Prussi, S. Ferrero, L. Oriani, P. Ottonello, P. Torre and F. Cherchi, *Biomass Bioenergy*, 2012, **46**, 25–35.

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

15 S. Dutta, S. De, M. I. Alam, M. M. Abu-Omar and B. Saha, *J. Catal.*, 2012, **288**, 8–15.
16 I.-J. Kuo, N. Suzuki, Y. Yamauchi and K. C.-W. Wu, *RSC Adv.*, 2013, **3**, 2028–2034.
17 T. Timell and N. Syracuse, *Wood Sci. Technol.*, 1967, **1**, 45–70.
18 B. C. Saha, *J. Ind. Microbiol. Biotechnol.*, 2003, **30**, 279–291.
19 J. Zakzeski, P. C. A. Bruijnincx, A. L. Jongerius and B. M. Weckhuysen, *Chem. Rev.*, 2010, **110**, 3552–3599.
20 M. M. Campbell and R. R. Sederoff, *Plant Physiol.*, 1996, **110**, 3.
21 P. Spaulding, *Mo. Bot. Gard. Annu. Rep.*, 1906, 41–58.
22 A. Boudet, C. Lapierre and J. Grima-Pettenati, *New Phytol.*, 1995, 203–236.
23 H. Wikberg and S. L. Maunu, *Carbohydr. Polym.*, 2004, **58**, 461–466.
24 O. Derkacheva and D. Sukhov, *Macromol. Symp.*, 2008, vol. 265, pp. 61–68.
25 G. E. Maciel, J. F. Haw, D. H. Smith, B. C. Gabrielson and G. R. Hatfield, *J. Agric. Food Chem.*, 1985, **33**, 185–191.
26 K. P. Kringstad and K. Lindström, *Environ. Sci. Technol.*, 1984, **18**, 236A–248A.
27 T. C. Acharjee, C. J. Coronella and V. R. Vasquez, *Bioresour. Technol.*, 2011, **102**, 4849–4854.
28 P. J. Kleppe, *Tappi*, 1970, **53**, 35–47.
29 J. Johakimu and J. Andrew, *BioResources*, 2013, **8**, 3490–3504.
30 C. Mendes, M. Carvalho, C. Baptista, J. Rocha, B. Soares and G. Sousa, *Food Bioprod. Process.*, 2009, **87**, 197–207.
31 B. D. Ribeiro, *Poster presentation at the 3rd International Symposium on Green Chemistry*, La Rochelle, France, 2015.
32 A. Foulon, O. Hansen, O. Haberstroh, H. Bailom, J. Sextl and H. Pauly, *Ber. Dtsch. Chem. Ges. A*, 1934, **67**, 1177–1199.
33 S. A. Rydholm, *et al.*, *Pulping processes*, 1965.
34 T. N. Kleinert, *US Pat.*, US 3585104 A, 1971.
35 A. Johansson, O. Aaltonen and P. Ylinen, *Biomass*, 1987, **13**, 45–65.
36 E. Muurinen, *Organosolv pulping: A review and distillation study related to peroxyacid pulping*, Oulun yliopisto, 2000, vol. 146.
37 X. Zhao, K. Cheng and D. Liu, *Appl. Microbiol. Biotechnol.*, 2009, **82**, 815–827.
38 F. Hermanutz, F. Gähr, E. Uerdingen, F. Meister and B. Kosan, *Macromol. Symp.*, 2008, vol. 262, pp. 23–27.
39 I. Adorjan, A. Potthast, T. Rosenau, H. Sixta and P. Kosma, *Cellulose*, 2005, **12**, 51–57.
40 P. Wasserscheid and T. Welton, *Ionic liquids in synthesis*, Wiley Online Library, 2008, vol. 1.
41 H. Olivier-Bourbigou, L. Magna and D. Morvan, *Appl. Catal., A*, 2010, **373**, 1–56.
42 D. R. MacFarlane, N. Tachikawa, M. Forsyth, J. M. Pringle, P. C. Howlett, G. D. Elliott, J. H. Davis, M. Watanabe, P. Simon and C. A. Angell, *Energy Environ. Sci.*, 2014, **7**, 232–250.
43 M. A. P. Martins, C. P. Frizzo, A. Z. Tier, D. N. Moreira, N. Zanatta and H. G. Bonacorso, *Chem. Rev.*, 2014, **114**, DOI: 10.1021/cr500106x.
44 J. P. Hallett and T. Welton, *Chem. Rev.*, 2011, **111**, 3508–3576.
45 D. Parmentier, S. J. Metz and M. C. Kroon, *Green Chem.*, 2013, **15**, 205–209.
46 A. P. Abbott, D. Boothby, G. Capper, D. L. Davies and R. K. Rasheed, *J. Am. Chem. Soc.*, 2004, **126**, 9142–9147.
47 Q. Zhang, K. D. O. Vigier, S. Royer and F. Jérôme, *Chem. Soc. Rev.*, 2012, **41**, 7108–7146.
48 E. L. Smith, A. P. Abbott and K. S. Ryder, *Chem. Rev.*, 2014, **114**, 11060–11082.
49 M. Francisco, A. van den Bruinhorst and M. C. Kroon, *Angew. Chem., Int. Ed.*, 2013, **52**, 3074–3085.
50 F. Jutz, J.-M. Andanson and A. Baiker, *Chem. Rev.*, 2011, **111**, 322–353.
51 H. G. Morrison, C. C. Sun and S. Neervannan, *Int. J. Pharm.*, 2009, **378**, 136–139.
52 I. Mamajanov, A. E. Engelhart, H. D. Bean and N. V. Hud, *Angew. Chem., Int. Ed.*, 2010, **49**, 6310–6314.
53 N. R. Rodriguez, A. S. González, P. M. Tijssen and M. C. Kroon, *Fluid Phase Equilib.*, 2015, **385**, 72–78.
54 N. R. Rodriguez, B. S. Molina and M. C. Kroon, *Fluid Phase Equilib.*, 2015, **394**, 71–82.
55 L. F. Zubeir, M. H. Lacroix and M. C. Kroon, *J. Phys. Chem. B*, 2014, **118**, 14429–14441.
56 Y. H. Choi, J. van Spronsen, Y. Dai, M. Verberne, F. Hollmann, I. W. C. E. Arends, G.-J. Witkamp and R. Verpoorte, *Plant Physiol.*, 2011, **156**, 1701–1705.
57 Y. Dai, J. van Spronsen, G.-J. Witkamp, R. Verpoorte and Y. H. Choi, *J. Nat. Prod.*, 2013, **76**, 2162–2173.
58 Y. Dai, G.-J. Witkamp, R. Verpoorte and Y. H. Choi, *Food Chem.*, 2015, **187**, 14–19.
59 B. Kirchner, *Ionic Liquids*, Springer-Verlag, New york, 2009.
60 N. V. Plechkova and K. R. Seddon, *Chem. Soc. Rev.*, 2008, **37**, 123–150.
61 T. L. Greaves and C. J. Drummond, *Chem. Rev.*, 2008, **108**, 206–237.
62 B. Clare, A. Sirwardana and D. R. MacFarlane, *Ionic Liquids*, Springer, 2010, pp. 1–40.
63 M. Pucheault and M. Vaultier, *Ionic Liquids*, Springer, 2010, pp. 83–126.
64 A. A. Torriero, *Electrochemistry in Ionic Liquids*, Fundamentals, Springer, 2015, vol. 1.
65 M. Francisco, A. van den Bruinhorst and M. C. Kroon, *Green Chem.*, 2012, **14**, 2153–2157.
66 D. V. Wagle, H. Zhao and G. A. Baker, *Acc. Chem. Res.*, 2014, **47**, 2299–2308.
67 A. P. Abbott, G. Capper, D. L. Davies, R. K. Rasheed and V. Tambyrajah, *Chem. Commun.*, 2003, 70–71.
68 Y. Dai, J. van Spronsen, G.-J. Witkamp, R. Verpoorte and Y. H. Choi, *Anal. Chim. Acta*, 2013, **766**, 61–68.
69 C. Florindo, F. S. Oliveira, L. P. N. Rebelo, A. M. Fernandes and I. M. Marrucho, *ACS Sustainable Chem. Eng.*, 2014, **2**, 2416–2425.
70 M. C. Gutiérrez, M. L. Ferrer, C. R. Mateo and F. del Monte, *Langmuir*, 2009, **25**, 5509–5515.
71 S. Aparicio, M. Atilhan and F. Karadas, *Ind. Eng. Chem. Res.*, 2010, **49**, 9580–9595.
72 A. Mohammad, *Green solvents II: properties and applications of ionic liquids*, Springer Science & Business Media, 2012, vol. 2.

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

73 S. Zhang, N. Sun, X. He, X. Lu and X. Zhang, *J. Phys. Chem. Ref. Data*, 2006, **35**, 1475–1517.
74 M. Yoshizawa, W. Xu and C. A. Angell, *J. Am. Chem. Soc.*, 2003, **125**, 15411–15419.
75 J. MSS Esperança, J. N. Canongia Lopes, M. Tariq, L. M. Santos, J. W. Magee and L. P. N. Rebelo, *J. Chem. Eng. Data*, 2009, **55**, 3–12.
76 J. D. Holbrey, W. M. Reichert, M. Nieuwenhuyzen, S. Johnson, K. R. Seddon and R. D. Rogers, *Chem. Commun.*, 2003, 1636–1637.
77 C. Hardacre, J. D. Holbrey, P. B. McCormac, S. J. McMath, M. Nieuwenhuyzen and K. R. Seddon, *J. Mater. Chem.*, 2001, **11**, 346–350.
78 W. A. Henderson, P. Fylstra, C. Hugh, P. C. Trulove and S. Parsons, *Phys. Chem. Chem. Phys.*, 2012, **14**, 16041–16046.
79 A. Paiva, R. Craveiro, I. Aroso, M. Martins, R. L. Reis and A. R. C. Duarte, *ACS Sustainable Chem. Eng.*, 2014, **2**, 1063–1071.
80 D. J. G. P. van Osch, L. F. Zubeir, A. van den Bruinhorst, M. A. A. Rocha and M. C. Kroon, *Green Chem.*, 2015, **17**, 4518–4521.
81 X. Baokou and M. Anouti, *J. Phys. Chem. C*, 2014, **119**, 970–979.
82 M. Francisco, A. van den Bruinhorst, L. F. Zubeir, C. J. Peters and M. C. Kroon, *Fluid Phase Equilib.*, 2013, **340**, 77–84.
83 P. M. Dean, J. M. Pringle and D. R. MacFarlane, *Phys. Chem. Chem. Phys.*, 2010, **12**, 9144–9153.
84 D. Rooney, J. Jacquemin and R. Gardas, *Ionic Liquids*, Springer, 2010, pp. 185–212.
85 C. P. Fredlake, J. M. Crosthwaite, D. G. Hert, S. N. Aki and J. F. Brennecke, *J. Chem. Eng. Data*, 2004, **49**, 954–964.
86 H. F. Almeida, H. Passos, J. A. Lopes-da-Silva, A. M. Fernandes, M. G. Freire and J. A. Coutinho, *J. Chem. Eng. Data*, 2012, **57**, 3005–3013.
87 K. Marsh, J. Boxall and R. Lichtenthaler, *Fluid Phase Equilib.*, 2004, **219**, 93–98.
88 A. Goodwin, K. Marsh and W. Wakeham, *Measurement of the thermodynamic properties of single phases*, Elsevier, 2003.
89 M. Tariq, J. Esperança, M. Soromenho, L. Rebelo and J. C. Lopes, *Phys. Chem. Chem. Phys.*, 2013, **15**, 10960–10970.
90 M. Kanakubo and K. R. Harris, *J. Chem. Eng. Data*, 2015, **60**, 1408–1418.
91 X. Wang, Y. Chi and T. Mu, *J. Mol. Liq.*, 2014, **193**, 262–266.
92 D. J. G. P. van Osch, D. Parmentier, C. H. J. T. Dietz, A. van den Bruinhorst, R. Tuinier and M. C. Kroon, *Chem. Commun.*, 2016, **52**, 11987–11990.
93 B. D. Ribeiro, C. Florindo, L. C. Iff, M. A. Coelho and I. M. Marrucho, *ACS Sustainable Chem. Eng.*, 2015, **3**, 2469–2477.
94 A. P. Abbott, G. Capper and S. Gray, *ChemPhysChem*, 2006, **7**, 803–806.
95 A. Hayyan, F. S. Mjalli, I. M. AlNashef, Y. M. Al-Wahaibi, T. Al-Wahaibi and M. A. Hashim, *J. Mol. Liq.*, 2013, **178**, 137–141.
96 K. Shahbaz, S. Baroutian, F. S. Mjalli, M. A. Hashim and I. M. AlNashef, *Thermochim. Acta*, 2012, **527**, 59–66.
97 L. Bahadori, M. H. Chakrabarti, F. S. Mjalli, I. M. AlNashef, N. S. A. Manan and M. A. Hashim, *Electrochim. Acta*, 2013, **113**, 205–211.
98 A. P. Abbott, R. C. Harris, K. S. Ryder, C. D'Agostino, L. F. Gladden and M. D. Mantle, *Green Chem.*, 2011, **13**, 82–90.
99 K. Shahbaz, F. S. Mjalli, M. A. Hashim and I. M. AlNashef, *Thermochim. Acta*, 2011, **515**, 67–72.
100 D. R. Lide, *CRC handbook of chemistry and physics*, CRC Press, 2004.
101 A. Bhattacharjee, P. J. Carvalho and J. A. Coutinho, *Fluid Phase Equilib.*, 2014, **375**, 80–88.
102 A. Bhattacharjee, J. A. Lopes-da-Silva, M. G. Freire, J. A. Coutinho and P. J. Carvalho, *Fluid Phase Equilib.*, 2015, **400**, 103–113.
103 M. A. Martins, C. M. Neves, K. A. Kurnia, P. J. Carvalho, M. A. Rocha, L. M. Santos, S. P. Pinho and M. G. Freire, *Fluid Phase Equilib.*, 2016, **407**, 188–196.
104 M. A. Martins, C. M. Neves, K. A. Kurnia, A. Luís, L. M. Santos, M. G. Freire, S. P. Pinho and J. A. Coutinho, *Fluid Phase Equilib.*, 2014, **375**, 161–167.
105 J. Jacquemin, P. Husson, A. A. H. Padua and V. Majer, *Green Chem.*, 2006, **8**, 172–180.
106 S. N. Baker, G. A. Baker and F. V. Bright, *Green Chem.*, 2002, **4**, 165–169.
107 J. A. Widegren, A. Laesecke and J. W. Magee, *Chem. Commun.*, 2005, 1610–1612.
108 S. Saha and H. Hamaguchi, *J. Phys. Chem. B*, 2006, **110**, 2777–2781.
109 A. P. Abbott, R. C. Harris and K. S. Ryder, *J. Phys. Chem. B*, 2007, **111**, 4910–4913.
110 D. Shah and F. S. Mjalli, *Phys. Chem. Chem. Phys.*, 2014, **16**, 23900–23907.
111 S. Matsuo and T. Makita, *Int. J. Thermophys.*, 1989, **10**, 833–843.
112 M. Tariq, P. J. Carvalho, J. A. Coutinho, I. M. Marrucho, J. N. C. Lopes and L. P. Rebelo, *Fluid Phase Equilib.*, 2011, **301**, 22–32.
113 D. Coleman and N. Gathergood, *Chem. Soc. Rev.*, 2010, **39**, 600–637.
114 N. Gathergood, P. J. Scammells and M. T. Garcia, *Green Chem.*, 2006, **8**, 156–160.
115 M. T. Garcia, N. Gathergood and P. J. Scammells, *Green Chem.*, 2005, **7**, 9–14.
116 N. Gathergood, M. T. Garcia and P. J. Scammells, *Green Chem.*, 2004, **6**, 166–175.
117 B. Kudłak, K. Owczarek and J. Namieśnik, *Environ. Sci. Pollut. Res.*, 2015, **22**, 11975–11992.
118 S. P. Ventura, R. L. de Barros, T. Sintra, C. M. Soares, A. S. Lima and J. A. Coutinho, *Ecotoxicol. Environ. Saf.*, 2012, **83**, 55–62.
119 M. Hayyan, M. A. Hashim, A. Hayyan, M. A. Al-Saadi, I. M. AlNashef, M. E. S. Mirghani and O. K. Saheed, *Chemosphere*, 2013, **90**, 2193–2195.
120 K. Radošević, M. C. Bubalo, V. G. Srček, D. Grgas, T. L. Dragičević and I. R. Redovniković, *Ecotoxicol. Environ. Saf.*, 2015, **112**, 46–53.
121 R. P. Swatloski, S. K. Spear, J. D. Holbrey and R. D. Rogers, *J. Am. Chem. Soc.*, 2002, **124**, 4974–4975.
122 Y. Fukaya, K. Hayashi, M. Wada and H. Ohno, *Green Chem.*, 2008, **10**, 44–46.

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

123 H. Zhao, G. A. Baker, Z. Song, O. Olubajo, T. Crittle and D. Peters, *Green Chem.*, 2008, **10**, 696–705.
124 R. C. Remsing, G. Hernandez, R. P. Swatloski, W. W. Massefski, R. D. Rogers and G. Moyna, *J. Phys. Chem. B*, 2008, **112**, 11071–11078.
125 H. Liu, K. L. Sale, B. M. Holmes, B. A. Simmons and S. Singh, *J. Phys. Chem. B*, 2010, **114**, 4293–4301.
126 A. Xu, J. Wang and H. Wang, *Green Chem.*, 2010, **12**, 268–275.
127 J.-M. Lee, S. Ruckes and J. M. Prausnitz, *J. Phys. Chem. B*, 2008, **112**, 1473–1476.
128 F. Huo, Z. Liu and W. Wang, *J. Phys. Chem. B*, 2013, **117**, 11780–11792.
129 K. M. Gupta, Z. Hu and J. Jiang, *RSC Adv.*, 2013, **3**, 4425–4433.
130 B. Lu, A. Xu and J. Wang, *Green Chem.*, 2014, **16**, 1326–1335.
131 R. C. Remsing, R. P. Swatloski, R. D. Rogers and G. Moyna, *Chem. Commun.*, 2006, 1271–1273.
132 Y. Zhao, X. Liu, J. Wang and S. Zhang, *ChemPhysChem*, 2012, **13**, 3126–3133.
133 J. Zhang, H. Zhang, J. Wu, J. Zhang, J. He and J. Xiang, *Phys. Chem. Chem. Phys.*, 2010, **12**, 1941–1947.
134 R. C. Remsing, I. D. Petrik, Z. Liu and G. Moyna, *Phys. Chem. Chem. Phys.*, 2010, **12**, 14827–14828.
135 J. Zhang, H. Zhang, J. Wu, J. Zhang, J. He and J. Xiang, *Phys. Chem. Chem. Phys.*, 2010, **12**, 14829–14830.
136 T. G. Youngs, J. D. Holbrey, M. Deetlefs, M. Nieuwenhuyzen, M. F. Costa Gomes and C. Hardacre, *ChemPhysChem*, 2006, **7**, 2279–2281.
137 T. Youngs, C. Hardacre and J. Holbrey, *J. Phys. Chem. B*, 2007, **111**, 13765–13774.
138 N. Novoselov, E. Sashina, V. Petrenko and M. Zaborsky, *Fibre Chem.*, 2007, **39**, 153–158.
139 B. Mostofian, J. C. Smith and X. Cheng, *Cellulose*, 2014, **21**, 983–997.
140 T. Heinze, S. Dorn, M. Schöbitz, T. Liebert, S. Köhler and F. Meister, *Macromol. Symp.*, 2008, vol. 262, pp. 8–22.
141 G. Ebner, S. Schiehser, A. Potthast and T. Rosenau, *Tetrahedron Lett.*, 2008, **49**, 7322–7324.
142 T. Erdmenger, C. Haensch, R. Hoogenboom and U. S. Schubert, *Macromol. Biosci.*, 2007, **7**, 440–445.
143 J. Vitz, T. Erdmenger, C. Haensch and U. S. Schubert, *Green Chem.*, 2009, **11**, 417–424.
144 P. Bonhote, A.-P. Dias, N. Papageorgiou, K. Kalyanasundaram and M. Grätzel, *Inorg. Chem.*, 1996, **35**, 1168–1178.
145 J. M. Crosthwaite, M. J. Muldoon, J. K. Dixon, J. L. Anderson and J. F. Brennecke, *J. Chem. Thermodyn.*, 2005, **37**, 559–568.
146 S. Zhang, X. Qi, X. Ma, L. Lu and Y. Deng, *J. Phys. Chem. B*, 2010, **114**, 3912–3920.
147 R. Rinaldi, *Chem. Commun.*, 2011, **47**, 511–513.
148 M. Gericke, T. Liebert, O. A. E. Seoud and T. Heinze, *Macromol. Mater. Eng.*, 2011, **296**, 483–493.
149 A. Xu, Y. Zhang, Y. Zhao and J. Wang, *Carbohydr. Polym.*, 2013, **92**, 540–544.
150 F. P. Burns, P.-A. Themens and K. Ghandi, *Compos. Interfaces*, 2014, **21**, 59–73.
151 O. Kuzmina, S. Jankowski, A. Fabiainska, E. Sashina and D. Wawro, *Cellul. Chem. Technol.*, 2014, **48**, 45–51.
152 J. Miao, H. Sun, Y. Yu, X. Song and L. Zhang, *RSC Adv.*, 2014, **4**, 36721–36724.
153 D. M. Rein, R. Khalfin, N. Szekely and Y. Cohen, *Carbohydr. Polym.*, 2014, **112**, 125–133.
154 J.-M. Andanson, E. Bordes, J. Devémy, F. Leroux, A. A. Pádua and M. F. C. Gomes, *Green Chem.*, 2014, **16**, 2528–2538.
155 S. Veliogly, X. Yao, J. Devemy, M. G. Ahunbay, S. B. Tantekin-Ersolmaz, A. Dequidt, M. F. Costa Gomes and A. A. Pádua, *J. Phys. Chem. B*, 2014, **118**, 14860–14869.
156 H. Zhang, J. Wu, J. Zhang and J. He, *Macromolecules*, 2005, **38**, 8272–8277.
157 L. K. J. Hauru, M. Hummel, A. W. T. King, I. A. Kilpeläinen and H. Sixta, *Biomacromolecules*, 2012, **13**, 2896–2905.
158 C. Olsson, A. Idström, L. Nordstierna and G. Westman, *Carbohydr. Polym.*, 2014, **99**, 438–446.
159 M. Mazza, D. A. Catana, C. Vaca-Garcia and C. Cecutti, *Celulose*, 2009, **16**, 207–215.
160 M. Iguchi, T. M. Aida, M. Watanabe and R. L. Smith, *Carbohydr. Polym.*, 2013, **92**, 651–658.
161 K. M. Gupta, Z. Hu and J. Jiang, *RSC Adv.*, 2013, **3**, 12794–12801.
162 A. Pinkert, K. N. Marsh and S. Pang, *Ind. Eng. Chem. Res.*, 2010, **49**, 11809–11813.
163 A. W. King, J. Asikkala, I. Mutikainen, P. Järvi and I. Kilpeläinen, *Angew. Chem.*, 2011, **123**, 6425–6429.
164 A. Parviainen, A. W. T. King, I. Mutikainen, M. Hummel, C. Selg, L. K. J. Hauru, H. Sixta and I. Kilpeläinen, *ChemSusChem*, 2013, **6**, 2161–2169.
165 M. Sharma, C. Mukesh, D. Mondal and K. Prasad, *RSC Adv.*, 2013, **3**, 18149–18155.
166 E. Zhou and H. Liu, *Asian J. Chem.*, 2014, **26**, 3626–3630.
167 E. C. Achinivu, R. M. Howard, G. Li, H. Gracz and W. A. Henderson, *Green Chem.*, 2014, **16**, 1114–1119.
168 Y. Pu, N. Jiang and A. J. Ragauskas, *J. Wood Chem. Technol.*, 2007, **27**, 23–33.
169 S. H. Lee, T. V. Doherty, R. J. Linhardt and J. S. Dordick, *Biotechnol. Bioeng.*, 2009, **102**, 1368–1376.
170 D. Fu and G. Mazza, *Bioresour. Technol.*, 2011, **102**, 7008–7011.
171 B. G. Janesko, *Phys. Chem. Chem. Phys.*, 2011, **13**, 11393–11401.
172 S. S. Tan, D. R. MacFarlane, J. Upfal, L. A. Edye, W. O. Doherty, A. F. Patti, J. M. Pringle and J. L. Scott, *Green Chem.*, 2009, **11**, 339–345.
173 W. E. Hart, J. B. Harper and L. Aldous, *Green Chem.*, 2015, **17**, 214–218.
174 I. Kilpeläinen, H. Xie, A. King, M. Granstrom, S. Heikkinen and D. S. Argyropoulos, *J. Agric. Food Chem.*, 2007, **55**, 9142–9148.
175 M. Zavrel, D. Bross, M. Funke, J. Büchs and A. C. Spiess, *Bioresour. Technol.*, 2009, **100**, 2580–2587.
176 Q. Wang, K. Chen, J. Li, G. Yang, S. Liu and J. Xu, *BioResources*, 2011, **6**, 3034–3043.
177 S. Konrad, M. Andrew, G. Charles M. and D. H. Vaughan, *Chem. Commun.*, 2004, 2178–2179.
178 M. Ab Rani, A. Brant, L. Crowhurst, A. Dolan, M. Lui, N. Hassan, J. Hallett, P. Hunt, H. Niedermeyer and J. Perez-Arlandis, others, *Phys. Chem. Chem. Phys.*, 2011, **13**, 16831–16840.
179 Y. Fukaya, A. Sugimoto and H. Ohno, *Biomacromolecules*, 2006, **7**, 3295–3297.

 This journal is © the Owner Societies 2017

Published on 02 December 2016. Downloaded by Westlake University on 11/3/2025 2:00:22 AM.

180 N. Muhammad, Z. Man, M. Azmi Bustam, M. I. Abdul Mutalib and S. Rafiq, *J. Ind. Eng. Chem.*, 2013, **19**, 207–214.
181 D. Fu, G. Mazza and Y. Tamaki, *J. Agric. Food Chem.*, 2010, **58**, 2915–2922.
182 I. Anugwom, V. Eta, P. Virtanen, P. Mäki-Arvela, M. Hedenström, M. Hummel, H. Sixta and J.-P. Mikkola, *ChemSusChem*, 2014, **7**, 1170–1176.
183 J. L. Anderson, J. Ding, T. Welton and D. W. Armstrong, *J. Am. Chem. Soc.*, 2002, **124**, 14247–14254.
184 L. Crowhurst, P. R. Mawdsley, J. M. Perez-Arlandis, P. A. Salter and T. Welton, *Phys. Chem. Chem. Phys.*, 2003, **5**, 2790–2794.
185 F.-Y. Du, X.-H. Xiao, X.-J. Luo and G.-K. Li, *Talanta*, 2009, **78**, 1177.
186 Y. Fan, M. Chen, C. Shentu, F. El-Sepai, K. Wang, Y. Zhu and M. Ye, *Anal. Chim. Acta*, 2009, **650**, 65–69.
187 M. Li, C. U. Pittman Jr. and T. Li, *Talanta*, 2009, **78**, 1364–1370.
188 X. Cao, X. Ye, Y. Lu, Y. Yu and W. Mo, *Anal. Chim. Acta*, 2009, **640**, 47–51.
189 A. Pinkert, D. F. Goeke, K. N. Marsh and S. Pang, *Green Chem.*, 2011, **13**, 3124–3136.
190 A. K. Kumar, B. S. Parikh and M. Pravakar, *Environ. Sci. Pollut. Res.*, 2015, 1–11.
191 A. Procentese, E. Johnson, V. Orr, A. G. Campanile, J. A. Wood, A. Marzocchella and L. Rehmann, *Bioresour. Technol.*, 2015, **192**, 31–36.
192 D. A. Fort, R. C. Remsing, R. P. Swatloski, P. Moyna, G. Moyna and R. D. Rogers, *Green Chem.*, 2007, **9**, 63–69.
193 N. Sun, M. Rahman, Y. Qin, M. L. Maxim, H. Rodríguez and R. D. Rogers, *Green Chem.*, 2009, **11**, 646–655.
194 A. Brandt, J. P. Hallett, D. J. Leak, R. J. Murphy and T. Welton, *Green Chem.*, 2010, **12**, 672–679.
195 X. Wang, H. Li, Y. Cao and Q. Tang, *Bioresour. Technol.*, 2011, **102**, 7959–7965.
196 J. Viell and W. Marquardt, *Holzforschung*, 2011, **65**, 519–525.
197 W. Li, N. Sun, B. Stoner, X. Jiang, X. Lu and R. D. Rogers, *Green Chem.*, 2011, **13**, 2038–2047.
198 K. M. Torr, K. T. Love, Ö. P. Çetinkol, L. A. Donaldson, A. George, B. M. Holmes and B. A. Simmons, *Green Chem.*, 2012, **14**, 778–787.
199 A. Casas, M. Alonso, M. Oliet, T. Santos and F. Rodriguez, *Carbohydr. Polym.*, 2013, **92**, 1946–1952.
200 A. Casas, M. Oliet, M. V. Alonso, T. M. Santos and F. Rodriguez, *Ind. Eng. Chem. Res.*, 2013, **52**, 3628–3636.
201 A. M. da Costa Lopes, K. G. João, D. F. Rubik, E. Bogel-Łukasik, L. C. Duarte, J. Andreaus and R. Bogel-Łukasik, *Bioresour. Technol.*, 2013, **142**, 198–208.
202 Y. Hamada, K. Yoshida, R. Asai, S. Hayase, T. Nokami, S. Izumi and T. Itoh, *Green Chem.*, 2013, **15**, 1863–1868.
203 L. Liu, M. Ju, W. Li and Q. Hou, *Carbohydr. Polym.*, 2013, **98**, 412–420.
204 L. K. Hauru, Y. Ma, M. Hummel, M. Alekhina, A. W. King, I. Kilpeläinen, P. A. Penttilä, R. Serimaa and H. Sixta, *RSC Adv.*, 2013, **3**, 16365–16373.
205 H. Wang, G. Gurau, S. V. Pingali, H. M. O'Neill, B. R. Evans, V. S. Urban, W. T. Heller and R. D. Rogers, *ACS Sustainable Chem. Eng.*, 2014, **2**, 1264–1269.
206 M. Montalbo-Lomboy and D. Grewell, *Ultrason. Sonochem.*, 2015, **22**, 588–599.
207 K. Wilpiszewska and T. Spychaj, *Carbohydr. Polym.*, 2011, **86**, 424–428.
208 H. Passos, M. G. Freire and J. A. Coutinho, *Green Chem.*, 2014, **16**, 4786–4815.
209 A. Biswas, R. Shogren, D. Stevenson, J. Willett and P. K. Bhowmik, *Carbohydr. Polym.*, 2006, **66**, 546–550.
210 A. P. Abbott, A. D. Ballantyne, J. P. Conde, K. S. Ryder and W. R. Wise, *Green Chem.*, 2012, **14**, 1302–1307.
211 E. Leroy, P. Decaen, P. Jacquet, G. Coativy, B. Pontoire, A.-L. Reguerre and D. Lourdin, *Green Chem.*, 2012, **14**, 3063–3066.
212 Q. Xu, Q. Wang and L. Liu, *J. Appl. Polym. Sci.*, 2008, **107**, 2704–2713.
213 A. Sankri, A. Arhaliass, I. Dez, A. C. Gaumont, Y. Grohens, D. Lourdin, I. Pillin, A. Rolland-Sabaté and E. Leroy, *Carbohydr. Polym.*, 2010, **82**, 256–263.
214 E. Leroy, P. Jacquet, G. Coativy, A. laure Reguerre and D. Lourdin, *Carbohydr. Polym.*, 2012, **89**, 955–963.
215 S. Ramesh, R. Shanti and E. Morris, *Carbohydr. Polym.*, 2012, **87**, 701–706.
216 Y. Dai, G.-J. Witkamp, R. Verpoorte and Y. H. Choi, *Anal. Chem.*, 2013, **85**, 6272–6278.
217 Y. Dai, R. Verpoorte and Y. H. Choi, *Food Chem.*, 2014, **159**, 116–121.
218 Q. Cui, X. Peng, X.-H. Yao, Z.-F. Wei, M. Luo, W. Wang, C.-J. Zhao, Y.-J. Fu and Y.-G. Zu, *Sep. Purif. Technol.*, 2015, **150**, 63–72.
219 M. W. Nam, J. Zhao, M. S. Lee, J. H. Jeong and J. Lee, *Green Chem.*, 2015, **17**, 1718–1727.
220 Z.-F. Wei, X.-Q. Wang, X. Peng, W. Wang, C.-J. Zhao, Y.-G. Zu and Y.-J. Fu, *Ind. Crops Prod.*, 2015, **63**, 175–181.
221 X.-L. Qi, X. Peng, Y.-Y. Huang, L. Li, Z.-F. Wei, Y.-G. Zu and Y.-J. Fu, *Ind. Crops Prod.*, 2015, **70**, 142–148.
222 X.-H. Yao, D.-Y. Zhang, M.-H. Duan, Q. Cui, W.-J. Xu, M. Luo, C.-Y. Li, Y.-G. Zu and Y.-J. Fu, *Sep. Purif. Technol.*, 2015, **149**, 116–123.
223 H. E. Park, B. Tang and K. H. Row, *Anal. Lett.*, 2014, **47**, 1476–1484.
224 M.-H. Duan, M. Luo, C.-J. Zhao, W. Wang, Y.-G. Zu, D.-Y. Zhang, X. Yao and Y.-J. Fu, *Sep. Purif. Technol.*, 2013, **107**, 26–36.
225 M. Tian and K. Row, *J. Anal. Chem.*, 2011, **66**, 580–585.
226 A. L. Theis, A. J. Waldack, S. M. Hansen and M. A. Jeannot, *Anal. Chem.*, 2001, **73**, 5651–5654.
227 B. Tang, W. Bi, H. Zhang and K. H. Row, *Chromatographia*, 2013, **77**, 373–377.
228 A. Spietelun, Ł. Marcinkowski, M. de la Guardia and J. Namieśnik, *Talanta*, 2014, **119**, 34–45.
229 B. D. Ribeiro, M. A. Z. Coelho and I. M. Marrucho, *Eur. Food Res. Technol.*, 2013, **237**, 965–975.
230 B. D. Ribeiro, M. A. Z. Coelho, L. P. N. Rebelo and I. M. Marrucho, *Ind. Eng. Chem. Res.*, 2013, **52**, 12146–12153.
231 H. Lin, Y. Zhang, M. Han and L. Yang, *Ultrason. Sonochem.*, 2013, **20**, 680–684.